

CAN-to-RF Telemetry System for Solar Racing Vehicles

Bachelor Thesis

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Abstract

Real-time telemetry is critical in solar racing, where security monitoring, race strategy and predictive maintenance depend on continuous data transmission between the solar car and the strategy car following it. Race locations are often harsh, remote and lack cellular coverage, so the link must be autonomous, low-power, and secure. This thesis presents the design and implementation of a dual-link telemetry unit for aCentaury Solar Racing. The system pairs a long-range LoRa downlink at 868/915 MHz with a 2.4 GHz FLRC uplink carrying about 30 kbit/s of CAN payload, both driven by an STM32L4A6 microcontroller. All over-the-air data traffic is encrypted and authenticated with AES-128 in counter mode and CMAC, which the on-chip accelerator computes in about 30 μ s per packet. The link-budget model predicts a theoretical range of roughly 2.8 km on the MHz link and 550 m on the GHz link. Field tests confirm an operating range of 520 m and that strong cryptography is compatible with real-time performance on a low-power microcontroller. An autonomous, end-to-end secured, medium-range CAN telemetry link can be realised within the power, size, and regulatory constraints of a solar race car using commodity hardware.

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List of Abbreviations

ACMA	Australian Communications and Media Authority	IV	Initialisation Vector
AES	Advanced Encryption Standard	LDRO	Low Data Rate Optimisation
BER	Bit Error Rate	LIFO	Last-In First-Out
BLE	Bluetooth Low Energy	LIPD	Low Interference Potential Device (AU)
BMS	Battery Management System	LoRa	Long Range (Semtech CSS modulation)
BMU	Battery Management Unit	LoS	Line of Sight
BoM	Bill of Materials	LTE	Long-Term Evolution
BW	Bandwidth	LV	Low Voltage
BWSC	Bridgestone World Solar Challenge	MCU	Microcontroller Unit
CAN	Controller Area Network	MIMO	Multiple Input Multiple Output
CBC	Cipher Block Chaining	MPC	Multipath Component
CMAC	Cipher-based Message Authentication Code	MPPT	Maximum Power Point Tracker
CR	Coding Rate	NACK	Negative Acknowledgement
CRC	Cyclic Redundancy Check	NIST	National Institute of Standards and Technology
CSS	Chirp Spread Spectrum	OFCOM	Federal Office of Communications (CH)
CTR	Counter (block cipher mode)	OSI	Open Systems Interconnection
dB	Decibel	PCB	Printed Circuit Board
dB_i	Decibel relative to an isotropic radiator	PID	Proportional-Integral-Derivative
dBm	Decibel relative to 1 mW	PRBS	Pseudo-Random Binary Sequence
DC	Direct Current	PRF	Pseudo-Random Function
DMA	Direct Memory Access	PSK	Pre-Shared Key
ECB	Electronic Codebook	RF	Radio Frequency
ECDH	Elliptic-Curve Diffie–Hellman	RFIC	Radio Frequency Integrated Circuit
EIRP	Equivalent Isotropically Radiated Power	RNG	Random Number Generator
EMI	Electromagnetic Interference	RP-SMA	Reverse-Polarity SMA Connector

ERP	Effective Radiated Power	RSSI	Received Signal Strength Indicator
ETSI	European Telecommunications Standards Institute	RTOS	Real-Time Operating System
EU	European Union	RX	Receiver
FEC	Forward Error Correction	SD	Secure Digital
FHSS	Frequency Hopping Spread Spectrum	SF	Spreading Factor
FIPS	Federal Information Processing Standards	SMA	SubMiniature version A RF Connector
FLRC	Fast Long-Range Communication	SNR	Signal-to-Noise Ratio
FM	Fading Margin	SoC	State of Charge
GCM	Galois/Counter Mode	SPI	Serial Peripheral Interface
GFSK	Gaussian Frequency-Shift Keying	SRD	Short-Range Device
GMAC	Galois Message Authentication Code	TLS	Transport Layer Security
GMSK	Gaussian Minimum-Shift Keying	TX	Transmitter
GNSS	Global Navigation Satellite System	U.FL	Miniature Coaxial RF Connector (Hirose)
HMAC	Hash-based Message Authentication Code	UART	Universal Asynchronous Receiver-Transmitter
IMU	Inertial Measurement Unit	VCU	Vehicle Control Unit
IoT	Internet of Things	VNA	Vector Network Analyser
ISM	Industrial, Scientific, and Medical	Wi-Fi	Wireless LAN technology (IEEE 802.11)
ISR	Interrupt Service Routine	WPA2	Wi-Fi Protected Access 2
ITU	International Telecommunication Union		

Introduction

1.1 Motivation

The Bridgestone World Solar Challenge (BWSC) is a 3000 km endurance race from Darwin to Adelaide, in which small solar-powered vehicles compete across the Australian outback. The finishing rate hovers around 50 %, and outcomes are decided by the performance and reliability of the race car and strategy. Supervision of the car depends on a continuous flow of information from the solar car to a strategy car following behind: battery temperature and State of Charge (SoC), photovoltaic input, motor power, position, and a long list of secondary signals. Without this data the team cannot adapt to weather, terrain, or competitor behaviour, and the BWSC regulations explicitly require the Battery Officer to monitor critical battery parameters in real time. Because the route runs through remote regions, the link cannot rely on cellular infrastructure such as Long-Term Evolution (LTE); it must be self-contained, low-power, robust against line-of-sight (LoS) variation, and legal in both Switzerland (development) and Australia (competition). Off-the-shelf wireless CAN links exist but none simultaneously meets the constraints on power, size, latency, and range that a solar race car imposes.

1.2 Problem Statement

aCentauri Solar Racing, a student-led team at ETH Zürich, has finished the BWSC twice and intends to compete for the podium in future editions. A reliable, custom telemetry solution is a prerequisite for this ambition. The difficulty comes from the combination of constraints rather than any single one: the unit must fit inside the carbon-fibre aeroshell, stay within a power budget compatible with a solar-powered vehicle, withstand the electromagnetic environment of a high-voltage drivetrain, remain legally compliant in both Switzerland and Australia, and cope with the rapid signal variations of an outdoor line-of-sight radio channel. The transmitted data also carries strategically valuable information, so the link must provide confidentiality, integrity, and authenticity against eavesdropping and active interference. The problem addressed by this thesis is therefore the design, implementation, and characterisation of a wireless telemetry unit that bridges the on-board Controller Area Network (CAN) of the solar car to a receiver in the strategy car, ingesting CAN traffic at line rate and delivering it over an autonomous, cryptographically protected radio link.

1.3 Objectives and Contributions

The objective of this thesis is to design, implement, and characterise a CAN-to-RF telemetry system that satisfies the requirements in Section 1.4. The work makes the following contributions.

- A dual-radio architecture combining a sub-GHz LoRa link for robust long-range downlink with a 2.4 GHz FLRC link for high-rate uplink, motivated by a regulatory and propagation analysis of both bands.
- An STM32L4-based hardware design integrating CAN ingest, dual radio front-ends, and a power budget below 0.1 W in operating mode.
- A firmware stack covering modulation control, packet assembly, packet-loss detection with selective retransmission, session establishment, and authenticated encryption of all over-the-air traffic.
- An empirical evaluation of the resulting link against the theoretical limits and against the manufacturer's specifications of the radio front-ends.

1.4 Requirements

The main performance criteria of a telemetry solution for solar racing are reliability, bandwidth, range, latency, power consumption, and privacy. Based on these criteria, the system's functional and non-functional requirements are defined as follows.

Table 1.1: System requirements for the CAN-to-RF telemetry unit.

ID	Category	Requirement	Priority
F-01	Electrical	The telemetry unit shall wirelessly transmit CAN bus data from the solar car to the strategy car.	Must
F-02	Electrical	The telemetry unit shall have a CAN interface, handle all CAN communication in the car independently, and process data for transmission.	Must
F-03	Electrical	The telemetry unit should be able to detect packet loss and request retransmission to correct for random loss on the RF link.	Should
F-04	Security	The telemetry unit shall provide confidentiality, integrity, and authenticity for all communication between the nodes.	Must
NF-01	Performance	The wireless link shall transfer a CAN bus of 300 frames per second (fps) over a range of 500 m line of sight (LoS) with an effective frame loss of less than 30 %.	Must
NF-02	Reliability	The telemetry unit shall automatically re-establish connection when it was lost due to a reboot of unit.	Must
NF-03	Physical	The telemetry unit shall have a slim and lightweight form factor that fits within the aeroshell of the current solar car.	Must
NF-04	Utility	The system should use a hardware and software stack similar to other modules designed by aCentauri to ensure compatibility.	Should
NF-05	Electrical	The telemetry unit should consume less than 0.1 W in operating mode.	Should
NF-06	Electrical	The telemetry unit should introduce less than 1 second of latency.	Should

1.5 Evaluation Approach

The system is evaluated in three stages, presented in Chapter 4. Electrical verification covers design-rule checks during PCB design, short and continuity tests on the bare board, and peripheral bring-up after com-

ponent assembly. Software testing is conducted in a controlled desktop setup with attenuators between the two boards, where all packets are expected to arrive; CAN messages with known content are injected and the received values are checked end-to-end. Field tests then characterise range, packet loss, and Received Signal Strength Indicator (RSSI) against the link budget derived in Chapter 2.

1.6 Outline of this Work

Chapter 2 provides the background needed to follow the design decisions: regulatory and physical limits of radio communication, the two modulation schemes used, the main challenges of high-frequency PCB design, and cryptography on resource-constrained microcontrollers, together with the project-specific context on solar racing, CAN, and the STM32L4 family. Chapter 3 describes the design of the telemetry unit, structured along an adapted OSI model from the physical layer up to the application layer. Chapter 4 presents the electrical, software, and field test results and compares them against the requirements in Table 1.1 and the theoretical predictions of Chapter 2. Chapter 5 summarises the outcome and discusses the trade-offs made, and Chapter 6 lists extensions that did not fit the project timeline. The attachments contain links to software, documentation, and hardware manufacturing files.

Background

2.1 Solar Racing

Solar racing is a sport in which small electric vehicles, powered only by solar cells on the surface of the car, race against each other. The sport had its debut in 1985 at the Tour de Sol in Switzerland; today the most prominent event is the Bridgestone World Solar Challenge (BWSC) in Australia. Most teams are student organisations backed by private sponsors or universities. The key performance indicators of a solar car are reliability, aerodynamics, photovoltaics, rolling resistance, weight, and power consumption; on race day strategy and weather analysis also matter. aCentauri Solar Racing is a young team formed mainly of ETH Zürich students that has finished the BWSC twice without seriously competing for the title yet.

2.2 Controller Area Network

The Controller Area Network (CAN) is the connection between low-voltage modules inside most race cars. The differential nature of CAN allows long transmission distances even under Electromagnetic Interference (EMI), which is significant in race cars where electrical components are scattered around a compact body. The nominal data rate of Silvretta, the latest solar car built by aCentauri, is 500 kbit/s, and empirical measurements show that the bus operates at about 5 to 10 % of its maximum load.

In a CAN multiple nodes communicate over a shared bus. Priority is assigned to messages rather than nodes: every message carries an identifier, and a lower-priority transmission can be interrupted by a higher-priority one. Each module uses a filter to receive only the identifiers relevant to it; to keep these consistent, the firmware of every microcontroller shares a common message tree from which all publishers and call-backs are generated. The low-voltage (LV) system of Silvretta contains around ten autonomous modules covering the Vehicle Control Unit, Steering Wheel, Logger, Telemetry, and the LV sides of the Inverter, Photovoltaic Disconnect, Maximum Power Point Trackers, and Battery Management System. The telemetry unit gathers messages from all of these; Table 1 in the appendix lists their responsibilities.

2.3 STM32 Microcontrollers

Reliability, availability, and low power consumption are the main criteria in chip selection for low-voltage modules. The STM32L4 family offers a wide range of connectivity options and is well documented. These microcontrollers traditionally run bare-metal, but a Real-Time Operating System (RTOS) can be used to handle multiple concurrent functionalities. Their limited compute and flash require resource-efficient firmware.

2.4 Radio Communication

Radio Frequency (RF) communication transmits data over electromagnetic waves. The encapsulated data is mostly digital, and the mapping of a digital symbol to an analog wave (and back) is called modulation/demodulation; modern schemes map multiple bits per symbol using several wave properties (Section 2.4.3).

An antenna picks up more than just the transmitted signal. Anything additional is RF noise [11], with three main sources: thermal noise from background radiation, spurious emissions from electrical appliances, and intentional emissions from other users of the same or a nearby band. Interference with other users is why most nations regulate the spectrum (Section 2.4.4). The Signal-to-Noise Ratio (SNR) at the receiver is primarily shaped by the TX power and the link budget; the resulting Bit Error Rate (BER) is further improved by signal-processing techniques. This project uses an asymmetric full-duplex RF connection in which the uplink from solar car to strategy car has higher data rate and the downlink has higher range.

2.4.1 Physical Limitations

RF channels combine two propagation effects that complicate range prediction [11]. **Small-scale fading** arises because the receiver picks up the Line-of-Sight (LoS) component together with reflected Multipath Components (MPCs) from Interacting Objects in the surroundings; depending on their relative phase the components add constructively or destructively, so centimetre-scale movements of the transmitter, receiver, or environment can produce large amplitude swings, especially at high carrier frequencies (Figure 2.1). **Large-scale fading** is the classical shadowing effect: when a non-radio-transparent object blocks the LoS signal, the transition between light and dark zones is gradual because the receiver still picks up reflected energy. The large number of MPCs makes the exact signal amplitude at a given receiver location almost impossible to predict from first principles.

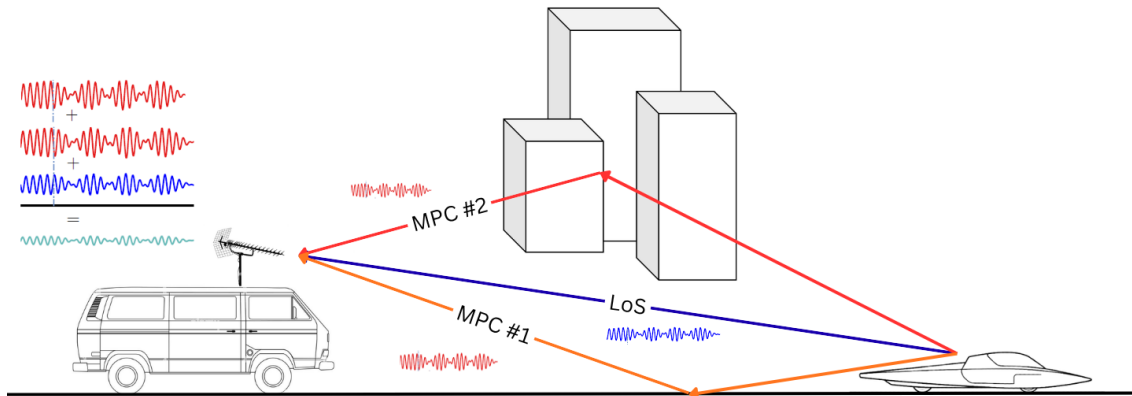


Figure 2.1: Small-scale fading from multipath propagation. The blue ray is the LoS component; the red and orange rays are reflections whose relative phase causes constructive or destructive addition at the receiver.

Link Budget

To estimate the TX power required for a given range, or to predict the achievable range, the link budget connects the TX power P_{TX} to the SNR at the receiver [11]. Most contributing factors enter multiplicatively,

so equations are written in logarithmic form: powers in dB relative to 1 mW (dBm) and antenna gains in dB relative to an isotropic radiator. Gains and losses then combine by addition.

The Equivalent Isotropically Radiated Power (EIRP) leaving the transmitter is

$$EIRP = P_{TX} + G_{TX} - L_{f,TX},$$

where $L_{f,TX}$ aggregates trace, connector, and combiner losses in the TX. The signal then suffers a path loss L_{path} during propagation. In a free-space environment, Friis' law gives a d^{-2} decay. Earth-bound RF additionally suffers from small-scale fading caused by the ground plane (Section 2.4.1); beyond a breakpoint distance $d_{break} = (4h_{TX}h_{RX})/\lambda_{f_c}$ a different path-loss function applies [11]:

$$L_{path} = \underbrace{32 + 20 \log_{10} \left(\frac{f_c}{1 \text{ GHz}} \right)}_{\text{Base loss at 1m (adjusted for } f_c)} + \underbrace{20 \log_{10}(d_{break})}_{\text{Free-space loss 1m to } d_{break}} + \underbrace{\alpha_{LBP} \cdot 10 \log_{10} \left(\frac{d}{d_{break}} \right)}_{\text{Rapid decay beyond } d_{break}}$$

with path-loss exponent α_{LBP} typically between 3.5 and 4.5. The received power is $P_{RX} = EIRP + G_{RX} - L_{f,RX} - L_{path}$, and for successful demodulation it must hold that $P_{RX} > P_{min}$. The minimum received power combines thermal noise, receiver noise figure F_{dB} , receiver bandwidth BW (over which the noise is collected), and the SNR required by the chosen modulation:

$$P_{min} = -174 \text{ dBm/Hz} + F_{dB} + 10 \cdot \log_{10}(BW) + SNR_{Req}.$$

In this work P_{min} is not evaluated from this expression but read directly from the datasheet receiver sensitivity of each transceiver, which already folds in the device noise figure and the SNR required by its modulation. No separate fading margin is added on top, so the ranges reported in Section 3.1.4 are the sensitivity-limited maxima ($P_{RX} = P_{min}$). Rearranging for the maximum range yields the closed-form expression

$$d = \frac{(4h_{TX}h_{RX})f_c}{c} \times 10^{\frac{EIRP - P_{Min} + G_{RX} - L_{f,RX} - 32 - 20 \log_{10}(f_c) - 20 \log_{10}(\frac{(4h_{TX}h_{RX})f_c}{c})}{10 \cdot \alpha_{LBP}}}$$

under the simplifying assumption that feeder losses and antenna gains are equivalent at TX and RX. Most parameters of this equation are fixed by physics or regulation; the handful that the designer can actually trade off (carrier frequency, bandwidth, antenna gain, feeder loss, and receiver noise figure) span the entire design space of an RF link. Project-specific values are derived in Section 3.1.4.

2.4.2 Electrical Challenges of Wireless Communication

High-frequency electronics differ from Direct Current (DC) or low-frequency design because every real capacitor, resistor, and inductor exhibits parasitic resistance, capacitance, and inductance that grow with frequency [3]. Two effects dominate the efficiency of the path from the RFIC to the antenna. **Insertion loss** is the power absorbed by any component placed between source and load; it is reduced by short interconnects, low-loss materials, and low-parasitic parts. **Impedance mismatch** causes power to be reflected back to the source; full power is delivered when the load impedance equals the complex conjugate of the source impedance [3]. This project uses commercial RF parts and antennas rated for the same frequency band, so the reactive part can be assumed correct by design, and the PCB-level work focuses on matching the real part to the system's characteristic impedance of 50Ω (Section 3.1.1). The standard tool to visualise impedance and reflection is the Smith chart, a polar plot of the reflection coefficient Γ ; a vector network analyser (VNA) using the Smith chart (Figure 4) was used to characterise antennas and RF components in this work (Section 4.1).

An antenna is a passive transducer between an electrical signal and an electromagnetic wave¹. Its shape determines a radiation pattern: omnidirectional antennas radiate equally in all directions and have no directional gain, while directional antennas concentrate energy into a specific direction, trading coverage angle for additional range (Section 3.1.2).

2.4.3 Modulation Techniques

Having looked at the physical factors that influence the success of an RF transmission, it is time to look at the techniques used to recover bits from a noisy signal. The theoretical background for the choices below follows the treatment in [11]. This project uses two different modulation schemes, one for each link direction. The downlink prioritises range and operates with LoRa in the sub-GHz band. The uplink prioritises throughput and uses FLRC at 2.4 GHz.

LoRa

LoRa is a proprietary physical-layer modulation developed by Semtech and based on Chirp Spread Spectrum (CSS). A chirp is a sinusoidal signal whose frequency sweeps linearly across a defined bandwidth; information is encoded by the frequency offset at which the sweep starts, so an unmodulated upchirp represents symbol value zero and all other symbols are shifted versions of it. A LoRa channel is described by its bandwidth BW and spreading factor SF , giving symbol duration $T_s = 2^{SF}/BW$ and $|S| = 2^{SF}$ distinct symbols carrying SF raw bits each. On the receiver side the incoming chirp is multiplied by the complex conjugate of an unmodulated upchirp, collapsing the sweep into a single tone whose frequency corresponds to the encoded symbol value, which is then recovered by a Discrete Fourier Transform. Because each chirp covers the full bandwidth, the modulation is robust against narrow-band interference, multipath effects, and Doppler shifts, and the scheme has low power consumption.

Above the raw symbol layer, LoRa wraps the payload in a short pipeline before transmission [16]: whitening (XOR with a PRBS), Hamming coding for redundancy, interleaving across symbols so that a corrupted symbol becomes single-bit errors rather than a burst, Gray coding so that a frequency-bin misread produces only one bit error, and finally CSS modulation. The explicit header is always encoded at the strongest coding rate 4/8 because its content is needed to decode the rest of the packet [12]². Together these mechanisms give LoRa a sensitivity of around -123 dBm to -137 dBm depending on SF and BW , enabling kilometre-range link budgets at sub-watt transmit power. The price is throughput: long on-air times and the legal duty-cycle caps restrict LoRa to low data rates.

FLRC

Fast Long-Range Communication (FLRC) is a relatively new modulation scheme supported by modern Semtech transceivers. It is intended to fill the gap between high-rate but short-range standards such as Bluetooth Low Energy (BLE) and the long-range but slow LoRa [4].

Public technical documentation on FLRC is significantly more limited than for LoRa. Semtech has not released a detailed specification, and, unlike for LoRa, no public reverse engineering of FLRC was available at the time of writing (May 2026)³. What is publicly known comes from the SX1280 datasheet and from Semtech application notes. At a high level, FLRC is a coherent Gaussian-shaped binary modulation similar in spirit to GMSK, paired with a convolutional code (typically rate 1/2) and a configurable preamble and

¹Active antennas integrate amplification at the radiating element; none are used in this work.

²Whitening is less critical in this project because the payload is already encrypted and therefore statistically random; the RFIC executes it at negligible cost regardless.

³Reverse engineering FLRC and analysing its security properties would be an interesting project for the future and could benefit the IoT community.

sync-word structure. Available bit rates range from approximately 260 kbit/s up to 1.3 Mbit/s, with quoted sensitivities on the order of -101 dBm at the lowest data rate. Compared to LoRa under similar conditions, FLRC trades roughly 15 to 20 dB of link budget for roughly two orders of magnitude in throughput [13].

2.4.4 Legal Limitations

The radio spectrum is a regulated shared resource. Any system operating without a dedicated licence must use one of the bands designated for unlicensed Industrial, Scientific and Medical (ISM) or Short-Range Device (SRD) use and comply with the limits on transmit power, occupied bandwidth, duty cycle, and channel access. The telemetry unit must operate both in Switzerland during development and in Australia during the BWSC; the two regions do not share identical rules, and the regulations therefore drive both the hardware selection and the link budget.

Table 2.1: Legally significant TX characteristics of the two radios used during development in this thesis.

Parameter	MHz Radio (SX1262)	GHz Radio (SX1280)
Frequency	869.4 MHz	2 450 MHz
Bandwidth	125 kHz	300 kHz
Conducted TX Power	22 dBm (~ 158 mW)	13 dBm (~ 20 mW)
EIRP	27 dBm (~ 500 mW)	10 dBm (~ 10 mW)

Spectrum Limitations

Sub-GHz SRD bands. In Europe, the LoRa-relevant band is 863–870 MHz, governed by ETSI EN 300 220 [9] and enforced nationally by OFCOM in Switzerland [2]. The historically dominant LoRaWAN sub-band 868.0–868.6 MHz allows $+14$ dBm ERP at a 1 % duty cycle (36 s per rolling hour), which is sufficient for sensor networks that dump data periodically. Empirical tests show that in noisy conditions roughly 5 % of packets in this work require retransmission, so the 1 % cap would be limiting. The 869.4–869.65 MHz sub-band, which permits $+27$ dBm ERP at a 10 % duty cycle, is therefore used instead. Australia uses a different allocation: the LIPD class licence covers 915–928 MHz with a $+30$ dBm EIRP ceiling and no mandatory duty-cycle restriction, so the radio exists in two breakout variants of the same SX1262, tuned for the EU and AU bands respectively.

2.4 GHz ISM band. The 2.400–2.4835 GHz band is harmonised across the ITU regions, with the same $+20$ dBm EIRP ceiling in Europe under ETSI EN 300 328 and in Australia under the ACMA LIPD class licence [8]. This regulatory uniformity is the main practical advantage of operating at 2.4 GHz: a single hardware and firmware configuration is legal anywhere the team races. The trade-offs are roughly 8 dB more free-space path loss than 900 MHz at the same distance, dense occupation by Wi-Fi and Bluetooth, and shrinking antenna geometries that make mounting and impedance matching considerably more sensitive.

EN 300 328 additionally imposes an EIRP spectral density limit of $+10$ dBm per MHz of occupied bandwidth for non-FHSS transmitters, to prevent a narrowband signal from concentrating all of its power into a small slice of the shared band. The FLRC uplink in this project occupies only 300 kHz, so the entire transmitted power falls within a single 1 MHz measurement window and the density limit binds before the absolute $+20$ dBm cap. The conducted TX power is set to the SX1280 maximum of $+13$ dBm and the TX antenna gain is kept around 3 dBi (Section 3.1.2), so that the resulting EIRP lands right around the $+10$ dBm density limit. Frequency Hopping Spread Spectrum (FHSS) would relax the density limit to $+20$ dBm per 100 kHz and additionally improve interference resilience, since a jammer that blocks one channel only affects the link for the brief instant when the hopping sequence lands on it. In this project the regulatory benefit of FHSS would

be the lifted density limit, which would let a more directional TX antenna push the EIRP closer to +20 dBm; the implementation is treated as experimental in Section 3.1.3.

The asymmetric architecture is a direct response to these trade-offs: LoRa at 868/915 MHz for robust long-range downlink and FLRC at 2.4 GHz for high-rate uplink anywhere in the world.

2.5 Cryptography in Low-Power RF-Based IoT

A wireless link is more exposed than a wired one: any receiver in range can observe the legitimate signal and any transmitter in range can inject onto the same channel. For a race environment with strategic payload, this motivates cryptographic protection on top of the radio link itself.

2.5.1 Threat Model

Four classes of attack are relevant: **eavesdropping**, where an attacker demodulates the transmissions with a software-defined radio (mitigated by encryption); **injection**, where crafted packets follow the legitimate modulation and framing in the hope of being accepted (mitigated by message authentication); **replay**, where unmodified legitimate packets are recorded and retransmitted later (mitigated by freshness, typically a monotonic counter or session-unique key); and **jamming**, a denial-of-service attack orthogonal to cryptography that can only be mitigated by frequency hopping, spread-spectrum modulation, or physical separation.

2.5.2 Symmetric Block Ciphers, Modes, and Key Exchange

The Advanced Encryption Standard (AES) is the dominant symmetric block cipher in modern systems (NIST FIPS 197 [10]). AES-128 is the common choice for embedded systems: a strong security margin, fast enough on small microcontrollers, and so widespread that many chip manufacturers offer hardware acceleration. Two modes of operation are used in this work. **Counter mode (CTR)** converts the block cipher into a stream cipher [6]: a counter combined with a nonce to form an Initialisation Vector (IV) is encrypted into a keystream block which is XORed with the plaintext. CTR is parallelisable, requires no padding, and preserves length; its central security requirement is that the IV must never repeat under the same key. **Cipher-based Message Authentication Code (CMAC)** produces a fixed-length tag over an arbitrary message under the same key [7, 14]; it is preferred over HMAC in constrained devices because it reuses the AES primitive already needed for encryption. The two modes are combined in an encrypt-then-MAC construction in this work: the payload is encrypted with AES-CTR and then a CMAC tag is computed over the header and the ciphertext, so that tampering is detected before decryption is attempted.

Symmetric cryptography requires both endpoints to share a secret key. **Asymmetric key exchange**, most commonly Elliptic-Curve Diffie-Hellman (ECDH), avoids embedding a long-term secret in the device but costs tens of milliseconds per Cortex-M4 point multiplication and requires defence against man-in-the-middle attacks. **Pre-shared key (PSK) exchange**, where a long-term key is provisioned out-of-band and session keys are derived from it together with fresh random material, is significantly simpler and is the approach taken here. The trade-off is that physical compromise of any device reveals the master key and therefore the keys of all past and future sessions [1].

2.5.3 AES on STM32L4

Some STM32L4 chips include a dedicated AES hardware accelerator that implements AES-128/256 in ECB, CBC, CTR, GCM, GMAC, and CMAC modes [15]. A single 128-bit block completes in roughly 200 clock

cycles (compared with several thousand for a software implementation), which at the 80 MHz core clock used here is about 2.5 μ s per block. Encrypting an 88-byte payload therefore takes on the order of 10 μ s and computing the CMAC tag a further 20 μ s, both negligible compared with the $\sim$ 8 ms FLRC on-air time. Two practical considerations apply under FreeRTOS: the peripheral is shared between several users on both radio links and must be serialised through a mutex, and its key and IV registers persist between operations, so a context switch must reload them. Both are addressed in [Section 3.4](#).

Approach

This chapter describes the design of the telemetry unit from the ground up, following an adapted OSI model that builds from the physical layer up to the application layer. The session and presentation layers of the standard model are merged into a single security layer and the network layer is omitted, since the system is a point-to-point link with no routing. Figure 3.1 shows the adapted stack and its correspondence to sections.

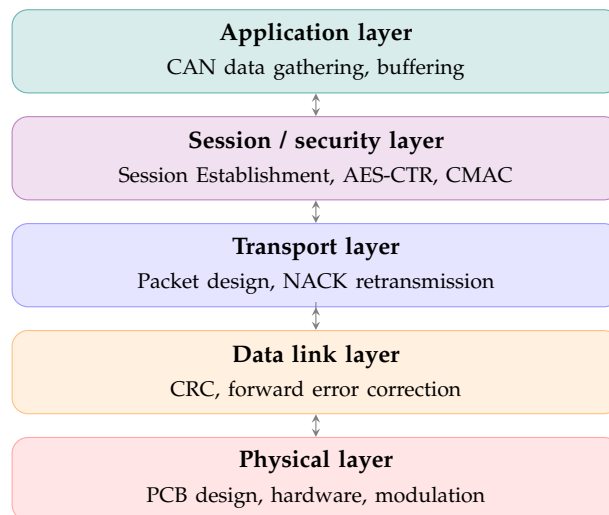


Figure 3.1: Adapted OSI model used to structure the design of the telemetry unit.

Table 3.1 maps the requirements from Table 1.1 onto the layers introduced above. Each requirement is addressed in the layer that has the most direct control over it. The mapping is referenced again from the relevant subsections in the rest of this chapter, so that the link between the high-level requirements and the concrete design choices stays explicit.

Table 3.1: Mapping of system requirements (Table 1.1) to the layers of the adapted OSI model in Figure 3.1.

Requirement	Layer	Section
F-01 (wireless transmission)	Physical	Section 3.1.2, Section 3.1.3
F-02 (CAN interface and processing)	Application	Section 3.5
F-03 (loss detection and retransmission)	Data Link, Transport	Section 3.2, Section 3.3.2
F-04 (confidentiality, integrity, authenticity)	Session / Security	Section 3.4
NF-01 (range, throughput, packet loss)	Physical, Data Link	Section 3.1.3, Section 3.1.4
NF-02 (auto re-connect)	Session / Security	Section 3.4.1
NF-03 (form factor)	Physical	Section 3.1.1, Section 3.1.2
NF-04 (compatibility with aCentauri stack)	Physical	Section 3.1.2
NF-05 (power consumption < 0.1 W)	Physical	Section 3.1.2
NF-06 (latency < 1 s)	Transport, Application	Section 3.3.2, Section 3.5

3.1 Physical Layer

The physical layer covers everything that determines how bits are turned into electromagnetic waves and back. In this project it encompasses the PCB layout and stackup, the choice of microcontroller and radio transceiver ICs, the antenna selection and matching, and the modulation parameters configured on each radio.

3.1.1 Printed Circuit Board Design

The hardware was designed in Altium Designer (the CAD tool used across aCentauri, which keeps libraries, footprints, and design rules consistent with the rest of the car) following the usual flow: requirements → component selection → schematic → layout → manufacturing → assembly → bring-up. The mechanical envelope matches the Raspberry Pi form factor (85 × 56 mm, three of four mounting holes, UART and 5 V connector positions) so that the board can stack directly on a Pi via the second STM32 UART; the Pi can then expose the decrypted telemetry stream over Wi-Fi or Ethernet, which is convenient for the strategy-car dashboard¹.

High-frequency traces are the main electrical difficulty here: every PCB feature between the RFIC and the antenna contributes parasitics that grow with frequency (Section 2.4.2), and the 50 Ω characteristic impedance must hold over the whole path. Two PCB iterations were produced. The first board routed the RF path as a controlled-impedance grounded coplanar waveguide on the top layer of a six-layer stack-up; VNA and RSSI measurements showed insertion and return losses noticeably worse than predicted by the stack-up calculator², and the link-budget targets were not met. The second board removes the printed RF path entirely: each RFIC breakout output is brought to a U.FL connector placed as close to the chip as practical, and a short pre-built shielded U.FL-to-SMA pigtail carries the signal to the chassis antenna connector. Pre-built RF cables are manufactured with tight impedance tolerances and full shielding, eliminating the PCB-level uncertainties at the cost of a little extra wiring inside the casing. The resulting board is shown in Figure 3.2 and a fully assembled unit in Figure 3.3; the full schematic of the second iteration is in Section 3.

Beyond the RF path, the board integrates a DC/DC buck converter feeding a clean 3.3 V rail for the MCU and both radio front-ends, with three power inputs: USB-C (negotiated to 5 V for development), a 5-pin M12

¹This stacking and the UART connection were tested successfully; the full Pi-side server, logger, and visualisation are left for future work.

²PCB manufacturers often do not enforce custom stack-ups in low-quantity orders.

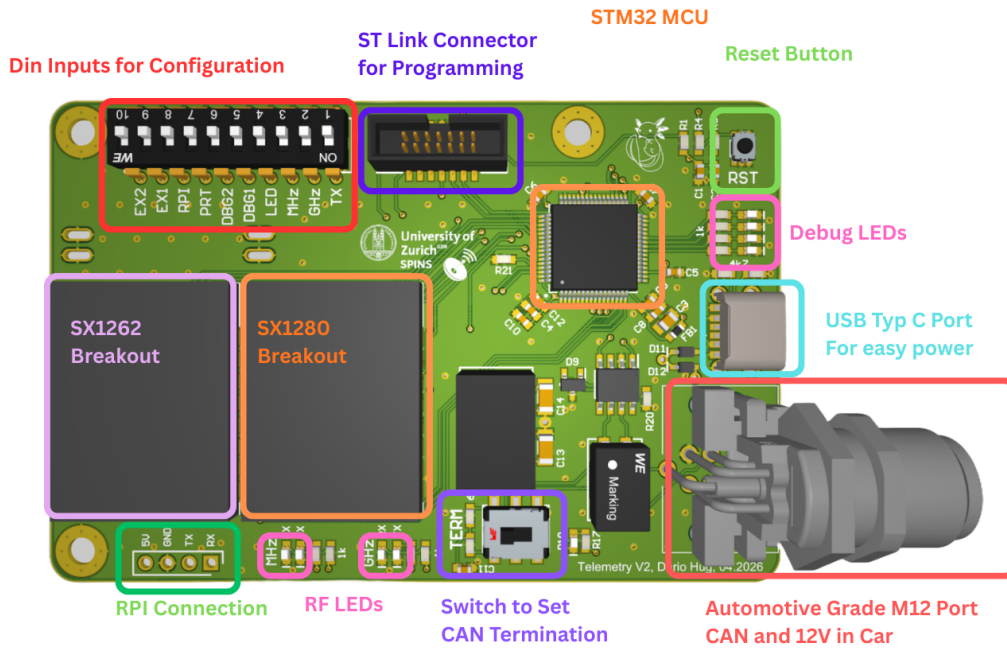


Figure 3.2: 3D rendering of the second iteration of the telemetry board, with the main components annotated.

connector that carries power and CAN in the car, and a Raspberry Pi header. Nine status LEDs (one power, eight firmware-controlled) signal link state, handshake progress, packet activity, and errors. A bank of ten Digital Input (DIN) switches allows hardware configuration without reflashing (board role, individual radio channels, LED patterns, debug verbosity, Raspberry Pi UART, FHSS mode), and a separate switch engages a $120\ \Omega$ termination between CAN high and low. A tactile reset button on the board edge allows a clean MCU reset during bench work.

3.1.2 Chosen Microcontrollers and Radio Hardware

The Microcontroller Unit (MCU) on both boards is an STMicroelectronics STM32L4A6RGT3, an Arm Cortex-M4F at up to 80 MHz with 1 MB flash and 320 kB SRAM [15]. The chip provides everything the telemetry unit needs: a CAN controller, three SPI peripherals (two for the radio front ends), a DMA controller, several UARTs for debug and host communication, and a true RNG for session nonces. The L4 family’s low-power architecture keeps the MCU subsystem well below the 0.1 W budget of NF-05 (Table 1.1) even at full clock. The rest of the aCentauri stack uses the STM32L496; the L4A6 is selected here because it is the smallest L4 variant that integrates the hardware AES-128/256 accelerator needed to encrypt every outgoing packet (Section 2.5.3, Section 3.4), while remaining pin- and footprint-compatible with the L496 so that the team’s PCB library and FreeRTOS port are shared (NF-04).

The two radio front ends are both Semtech transceivers chosen to match the two regulatory regimes of Section 2.4.4. The sub-GHz link uses an SX1262, which supports LoRa and (G)FSK between 150 and 960 MHz at up to +22 dBm conducted output. The 2.4 GHz link uses an SX1280 [13], which supports LoRa, FLRC, GFSK and BLE-compatible modes at up to +13 dBm. Both chips expose a similar software model (SPI command interface, packet engine with hardware CRC and FEC, configurable IRQs, on-chip buffer) and are

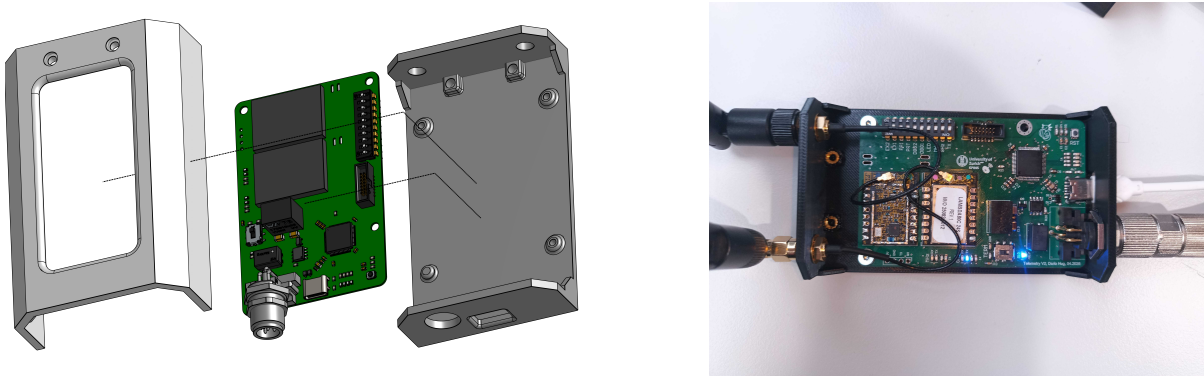


Figure 3.3: Telemetry unit in its 3D printed casing. Left: exploded view of the casing and board assembly. Final dimensions are 106x68x28 mm. Right: fully assembled unit with both antennas connected; power is supplied over USB-C and a Peak CAN USB interface generates the test CAN messages.

driven from independent SPI buses so that an in-flight uplink cannot stall an incoming downlink.

Antenna Selection

The fundamental constraint on antenna selection is frequency matching. An antenna is resonant only over a limited band, and operating it off-resonance raises its impedance, increases reflected power, and reduces the effective radiated power. A vector network analyser (VNA) was therefore used to measure the reflection coefficient of each candidate antenna before it was accepted into the link budget. Figure 3.4 shows the full set of antennas selected with the VNA setup.



Figure 3.4: Candidate antennas and the VNA used to characterise them. Left to right: two sub-GHz monopoles, 2.4 GHz whip antenna for RX, 2.4 GHz monopole antenna for TX.

MHz antennas. For the sub-GHz LoRa link a matched pair of slightly directional antennas is used at both TX and RX. The link budget margin is generous enough that the antenna choice is not critical here; the antennas on hand had an optimal tune around 890 MHz, close enough to 869 MHz to keep the reflection coefficient low. Field tests confirmed that the MHz link never approached its range limit under any test condition.

GHz antennas. The 2.4 GHz link calls for a more careful approach. As shown in Section 2.4.4, the EIRP spectral density constraint of EN 300 328 limits the effective radiated power to approximately +10 dBm. The EIRP is set entirely by the transmit chain: $EIRP = P_{TX} + G_{TX} - L_{f,TX}$ (Section 3.1.4). Antenna gain on the RX side, G_{RX} , enters the received power P_{RX} directly without increasing the radiated power. For this reason an asymmetric setup makes sense. A more directive RX antenna therefore improves the link budget without any regulatory consequence. The RX side on the GHz link is therefore equipped with a magnetic-mount whip antenna with a gain of roughly 14 dBi, while the GHz TX uses a monopole through-hole antenna with a gain of around 3 dBi, kept deliberately low to stay inside the EIRP density cap at full conducted output.

VNA measurements of the monopole antennas show that reflection of power is minimised, and consequently gain and impedance match are best, at approximately 2.45 GHz rather than at the originally planned 2.404 GHz. To operate the antenna at its resonant frequency, the centre frequency of the GHz radio is set to 2.45 GHz. The trade-off is a modest increase in Wi-Fi interference: 2.45 GHz sits in the middle of the 2.4 GHz Wi-Fi band, whereas 2.404 GHz lies below the lowest commonly used Wi-Fi channel and sees significantly less contention. In a race environment this is an acceptable compromise, because the link is mobile and any given interferer is only briefly encountered.

3.1.3 Modulation Techniques and Chosen Parameters

Both radios are configured at boot via SPI register writes and stay fixed during operation. Table 2.1 summarises the legal envelope.

LoRa in the MHz Range

The SX1262 is configured as $SF7$, $BW = 125$ kHz, $CR = 4/5$, with an 8-symbol preamble and hardware CRC enabled. Increasing SF improves sensitivity by roughly 2.5 dB per step but doubles the on-air time: at $SF7$ a 48-byte packet occupies the channel for ≈ 97 ms; at $SF8$ it exceeds 200 ms. Given the 10 % duty-cycle ceiling of the 869.4–869.65 MHz sub-band (Section 2.4.4) and that the downlink also carries NACKs under degraded conditions, $SF7$ leaves more headroom for retransmissions; the link-budget loss is modest, with the SX1262 sensitivity of -123 dBm at $SF7/BW = 125$ kHz still placing the predicted range (Section 3.1.4) well above the 500 m required by NF-01. $BW = 125$ kHz is the narrowest standard LoRa bandwidth that fits inside the 250 kHz sub-band without overlap, and $CR = 4/5$ adds 20 % redundancy without doubling the on-air time as $CR = 4/8$ would. The 8-symbol preamble is the minimum recommended for stable synchronisation, and Low Data Rate Optimisation is disabled because the symbol time of 1.024 ms is well below the 16 ms threshold at which LDRO becomes mandatory. Conducted TX power is at the SX1262 maximum of +22 dBm with a 200 μ s ramp; in the European variant the 9 dBi antenna gain and cable losses fully use the +27 dBm ERP allowance. The Australian variant only retunes the centre frequency to a channel in the LIPD 915–928 MHz allocation; all modulation parameters and the TX power are unchanged.

FLRC in the GHz Range

The SX1280 is configured for FLRC at 260 kbit/s with a 0.3 MHz channel bandwidth, coding rate 1/2, Gaussian shaping $BT = 0.5$, a 32-bit preamble, a 4-byte sync word, a 3-byte CRC, and whitening disabled. The centre frequency is set to 2.45 GHz to align with the resonant frequency of the chosen 2.4 GHz antennas (Section 3.1.2). FLRC offers bit rates from 260 kbit/s up to 1.3 Mbit/s; even at the lowest setting it is roughly two orders of magnitude faster than the LoRa downlink, more than enough for Silvertta's CAN traffic, so the lowest rate is selected to gain ≈ 9 dB of link budget and bring sensitivity close to -101 dBm. Coding rate 1/2 is the strongest FLRC FEC and recovers single-symbol losses that would otherwise become full-packet losses; whitening is disabled because the payload is already AES-encrypted and therefore statistically white. The 0.3 MHz bandwidth is the critical parameter for legal operation: because it is narrower than the 1 MHz measurement window of EN 300 328, the full transmitted power falls inside that window

and the +10 dBm/MHz spectral-density limit binds before the +20 dBm EIRP cap (Section 2.4.4). Conducted TX power is set to the SX1280 maximum of +13 dBm; combined with the 3 dBi TX antenna and the cable losses, the resulting EIRP lands right at the density limit.

Experimental: FHSS on the GHz Link

Operating FLRC on a fixed 2.45 GHz channel constrains the effective EIRP to $\approx +10$ dBm under EN 300 328 (Section 2.4.4). FHSS would relax this constraint and additionally improve interference resilience. In the experimental configuration the SX1280 is moved to a new channel after every transmission; after the initial handshake (which still runs on the fixed channel for ease of synchronisation) both endpoints follow the same deterministic hop sequence. The conducted output stays at +13 dBm; the regulatory benefit lies in lifting the density cap, which would allow a more directional TX antenna to push EIRP up to the full +20 dBm allowance. The remaining FLRC parameters are unchanged.

3.1.4 Resulting Theoretical Bandwidth and Range

With the hardware fixed in Section 3.1.2 and the modulation parameters fixed in Section 3.1.3, the closed-form range expression at the end of Section 2.4.3 can be evaluated with concrete numbers. Two simplifications are applied: both modules are assumed to be in free air (the carbon-fibre aeroshell of Silvretta is RF-blocking, but a fibre-glass canopy window behind which the antennas are mounted is RF-transparent, see Section 4.1), and the sub-GHz link is evaluated at 900 MHz as an approximation to the two regional variants (Europe at 869.4 MHz, Australia inside the LIPD 915 MHz allocation, see Section 2.4.4). The 5 % frequency difference has a negligible effect on the path loss at the distances of interest.

Table 3.2 collects the parameter values for both radios; TX power, bandwidth, and sensitivity follow from Section 3.1.3 and Table 2.1, antenna gains are measured for the antennas selected in Section 3.1.2, and feeder losses are approximated with some headroom and the path-loss exponent is a conservative value from the literature [11]. Consistent with Section 2.4.1, P_{Min} is the datasheet receiver sensitivity. The TX and RX antenna heights are interchanged between the two columns because the MHz link runs in the reverse direction, from the strategy car back to the solar car. The result is a theoretical range of 552 m at 2.4 GHz³ and 2 825 m at 900 MHz.

Both numbers exceed the 500 m line-of-sight requirement of NF-01. The asymmetry between the two bands is the quantitative motivation for the dual-link architecture. The empirical field measurements in Chapter 4 compare against these predictions.

³If FHSS is enabled, the EN 300 328 spectral-density cap on EIRP is lifted (see Section 3.1.3). The conducted output stays at the SX1280 maximum of +13 dBm, but a more directional TX antenna can push the EIRP above +10 dBm; the corresponding range scales with the additional gain.

Table 3.2: Link budget parameters and resulting theoretical range for both radios of the telemetry unit.

Symbol	Name	Description	Value 2.4 GHz	Value 900 MHz
h_{TX}	TX Height	Physical height of the transmit antenna above the ground plane.	0.6 m	2.5 m
h_{RX}	RX Height	Physical height of the receive antenna above the ground plane.	2.5 m	0.6 m
f_c	Carrier Frequency	Signal centre frequency.	2 450 MHz	900 MHz
c	Speed of Light	Propagation speed of electromagnetic waves in free space.	3×10^8 m/s	
$EIRP$	EIRP	Equivalent Isotropically Radiated Power.	10 dBm	27 dBm
P_{TX}	TX Power	Conducted output power of the transmitter amplifier.	13 dBm	22 dBm
$G_{Ant_{TX}}$	TX Antenna Gain	Gain of the TX antennas relative to an isotropic radiator.	3 dBi	9 dBi
$G_{Ant_{RX}}$	RX Antenna Gain	Gain of the RX antenna relative to an isotropic radiator.	14 dBi	9 dBi
$L_{f_{TX}}$	TX Feeder Loss	Signal attenuation from cables, connectors, and combiners in TX. (RP-SMA Antenna in the GHz TX needs an extra connector.)	6 dB	4 dB
$L_{f_{RX}}$	RX Feeder Loss	Signal attenuation from cables, connectors, and combiners in RX.	4 dB	4 dB
α_{LBP}	Path Loss Exponent	Decay rate in the log-distance model beyond the breakpoint.	4.5	4.5
d_{break}	Breakpoint Distance	Distance at which the signal transitions to rapid decay.	49 m	18 m
P_{Min}	RX Sensitivity	Datasheet receiver sensitivity for the chosen modulation; already includes the receiver noise figure and the required SNR.	-101 dBm	-123 dBm
BW	Bandwidth	Signal bandwidth determining total collected noise.	300 kHz	125 kHz
d	Range	Resulting theoretical TX/RX separation.	552 m	2 825 m

A visualisation of the link budget over distance for both radios is provided in Figure 3 in the appendix.

3.2 Data Link Layer

The data-link layer ensures that a frame leaving the transmitter arrives at the receiver as the same sequence of bits. It uses a Cyclic Redundancy Check (CRC) to detect that a frame was corrupted, and Forward Error Correction (FEC) to repair small numbers of bit errors before they reach the upper layers. Neither mechanism is implemented in software: both the SX1262 and the SX1280 contain a packet engine that handles CRC and FEC entirely in hardware as part of the modulation pipeline. The firmware sets a few configuration bytes during initialisation (Section 3.1.3) and then receives a clean payload through SPI, with frames that fail the CRC silently dropped by the chip.

Each radio adds its own error protection. On the MHz link the SX1262 guards every LoRa frame with a 16-bit CRC and a light error-correcting code, which together make bit errors rare. On the GHz link the SX1280 uses a wider 24-bit CRC and a stronger error-correcting code, because the 2.4 GHz band is noisier and runs at a higher bit rate. The stronger code costs some throughput, but the FLRC rate offers enough link-budget headroom to absorb it (Section 3.1.3). Together with the AES-CMAC tag added at the session layer, the chance that a corrupted packet slips through undetected is negligible for telemetry. From the upper layers'

perspective the data-link layer simply delivers either a clean frame or no frame at all; recovery from missing frames is the responsibility of the transport layer.

3.3 Transport Layer

The transport layer sits between the radio hardware and the application. It packages CAN data into units that fit on the air, numbers them for reordering, and recovers the ones lost in transit. Below it the radio delivers either a clean frame or no frame at all (Section 3.2); above it the application hands in CAN frames and assumes in-order delivery. The link is strongly asymmetric, so the layer defines one large packet format for the GHz side and one small packet format for the MHz side, and uses the MHz channel exclusively to signal that a GHz packet went missing.

3.3.1 Packet Design

All packets share the same skeleton: a fixed-size plaintext header, an AES-CTR-encrypted payload, and a truncated AES-CMAC tag (the cryptographic construction is described in Section 3.4). The GHz packet is 112 bytes long (Figure 3.5). The first eight bytes carry a 64-bit packet counter in plaintext, which doubles as the unique sequence number and as the per-packet AES-CTR IV under the session key. Byte 8 is the packet type; bytes 9–15 are reserved; bytes 16–103 are the encrypted payload; bytes 104–111 are the 8-byte CMAC tag computed over the header and ciphertext.

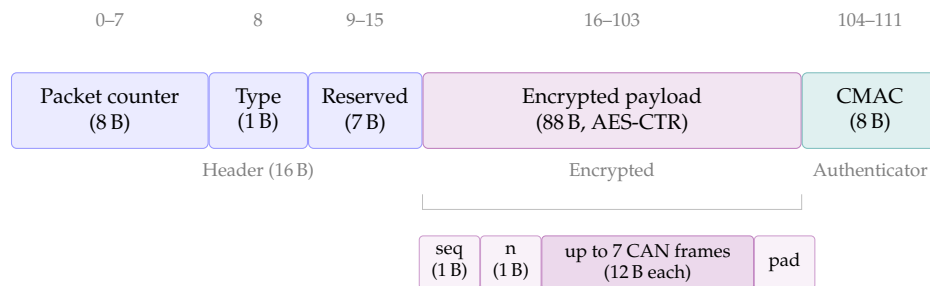


Figure 3.5: Layout of a GHz data packet (112 B). The header and the CMAC tag are in plaintext. The 88-byte payload is encrypted with AES-CTR and carries up to seven 12-byte CAN frames preceded by a one-byte sequence check and a one-byte frame count.

The encrypted payload has a fixed length regardless of how many CAN frames it carries; pacing is managed by the application layer. The first byte is a copy of the lowest byte of the packet counter, a cheap sanity check after decryption (with the right key it must match the header). The second byte gives the number of CAN frames in the packet (0–7). Each frame takes 12 bytes (4-byte identifier + 8-byte data), so up to 84 bytes of CAN content fit; unused bytes are zero-padded before encryption. The MHz packet uses the same skeleton but is only 64 bytes long, with a 40-byte encrypted payload that carries up to nine 4-byte counter values to retransmit. The two formats are summarised in Table 3.3.

The type byte in the header tells the receiver how to interpret the payload. The transport layer uses five values:

- **DATA** (GHz): the payload contains one or more CAN frames. This is the common case during operation.

Table 3.3: Summary of the two packet formats. The header and CMAC are identical on both links; only the payload differs.

Field	GHz packet	MHz packet
Total size	112 B	64 B
Header	16 B	16 B
Encrypted payload	88 B	40 B
CMAC tag	8 B	8 B
Max payload entries	7 CAN frames	9 NACK counters

- **ALL_EMPTY** (GHz): the payload contains no CAN frames. The transmitter still sends one such packet per second as a keepalive, so that the receiver can confirm the link is alive when there is no CAN.
- **RETX** (GHz): a copy of an earlier DATA packet, resent in response to a NACK. The plaintext counter is the original one, which is what the receiver needs to put the packet back into its sequence.
- **NACK** (MHz): the payload lists up to nine counter values that the receiver is still missing.
- **ALL_GOOD** (MHz): reserved for explicit "no gaps" status messages. The current firmware stays silent when the link is healthy and never sends this type, but the slot is kept open for a future bidirectional health-monitoring scheme.

Two further packet types, **HELLO** and **AUTH**, exist but belong to the handshake described at the session layer (Section 3.4). They reuse the same header skeleton so that the radio task does not need to know whether the packet it is transmitting is transport-layer or session-layer traffic.

3.3.2 Retransmissions

A wireless link occasionally loses packets even with the CRC and FEC of Section 3.2 doing their job; F-03 requires the transport layer to recover from this loss. The transport layer implements this with a counter-gap detector on the RX side, a NACK message back over the MHz channel, and a circular retransmission buffer on the TX side (Figure 3.6).

Gap detection. The receiver keeps a single *expected counter* variable. When a GHz packet passes the CMAC check, the receiver compares its counter to this expectation: a match increments the expected value and delivers the packet to the application; a higher counter records every value in between as missing and jumps the expected value to one above the received counter. Up to 64 missing counters can be tracked at once – about 1.3 seconds of complete blackout at the typical packet rate.

NACK transmission. A dedicated task on the RX wakes periodically, queries the gap tracker, and, if there are any gaps, ships a NACK packet listing over the MHz radio. Each missing counter is attempted up to five times across consecutive polls; if all five fail, the entry is dropped and the packet is considered permanently lost.

Retransmission buffer. On the TX side a circular buffer holds the last 512 GHz packets (roughly 10 seconds of CAN traffic), indexed by counter. When a NACK arrives, the transmitter looks up each requested counter, patches the type byte from DATA (or ALL_EMPTY) to RETX, recomputes the CMAC tag (the type byte is part of the input), and queues the packet. The encrypted payload is never recomputed: the AES-CTR keystream depends on the counter, which is unchanged, so a retransmission is cheap in compute cost. The mechanism consumes $512 \times 112 = 57 \text{ kB}$ of the 320 kB SRAM of the STM32L4A6 (Section 3.1.2).

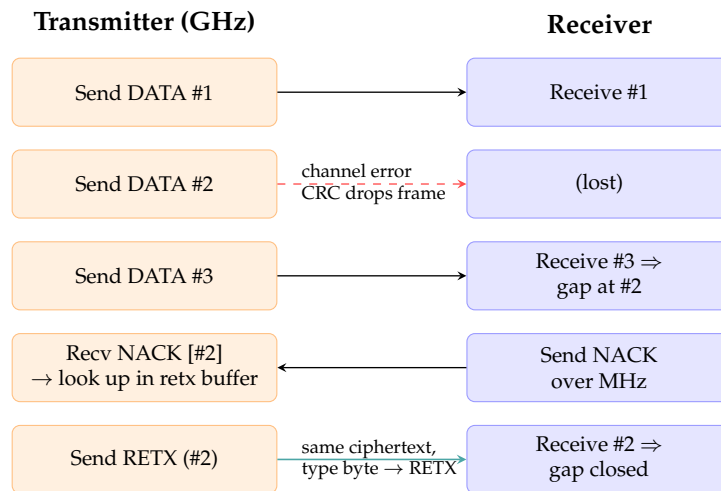


Figure 3.6: Retransmission flow. Packet #2 is lost on the GHz link; the receiver detects the counter gap, sends a NACK over the MHz link, and the transmitter resends the stored ciphertext with a RETX type byte.

3.4 Session / Security Layer

The session layer meets requirements F-04 (confidentiality, integrity, and authenticity) and NF-02 (automatic re-establishment after connection loss). The mapping to the threats of Section 2.5.1 is direct: eavesdropping is countered with AES-CTR encryption, injection with a CMAC tag on every packet, replay with a per-session key combined with a strictly monotonic packet counter, and peer reboots with a freshness-checked HELLO message. Jamming cannot be solved cryptographically and is left to the FHSS experiment of Section 3.1.3. The layer runs in two phases: a short post-boot handshake derives a 128-bit session key from a pre-shared master key and two fresh nonces, after which every outgoing packet is encrypted with AES-128-CTR under the session key and authenticated with a truncated AES-CMAC tag; the same code paths handle decryption, MAC verification, and replay rejection on the receiving side. Figure 3.7 shows the lifecycle.

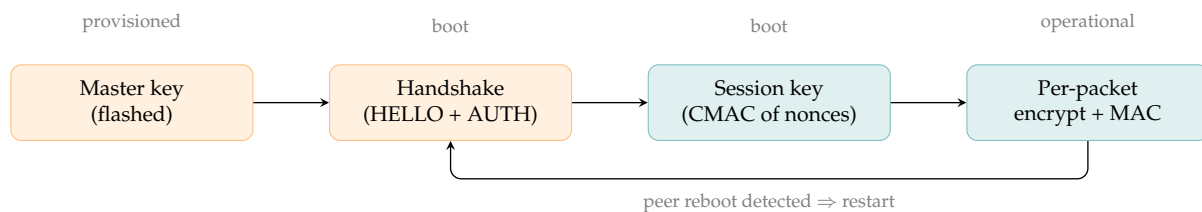


Figure 3.7: Lifecycle of the security layer. The master key is provisioned once at programming time. After boot the two modules run a handshake to derive a session key, then switch to per-packet encryption and authentication. A peer reboot detected during operation triggers a fresh handshake.

3.4.1 Master Key and Session Establishment

Master key and nonces. Both modules share a 128-bit master key K_M , generated by an offline tool and embedded into the firmware image as a constant; the key is present on every flashed unit but never appears on the radio. This is the PSK model of Section 2.5.2, chosen for its simplicity over an asymmetric exchange. Each module additionally generates a fresh 128-bit nonce at boot from the STM32L4A6 hardware RNG (seeded from on-chip thermal noise); nonces are kept in volatile memory only.

Handshake messages. Two packet types are reserved on top of the layout in Section 3.3.2. **HELLO** carries the local nonce and an 8-byte master-key proof $\tau_H = \text{CMAC}(K_M, \text{"HSv1HELLO"} \parallel N_{\text{local}})_{[0..7]}$, where the leading 8-byte ASCII string is a domain-separation tag [1] that prevents a HELLO proof from being mistaken for an AUTH proof. Because an in-session HELLO triggers a re-handshake, this proof of knowledge of K_M is necessary. **AUTH** carries the local nonce (so the peer can recover it if its HELLO was lost) and a full 16-byte CMAC proof $\tau_A = \text{CMAC}(K_M, N_{\text{peer}})$. Computing τ_A requires knowledge of K_M and possession of the freshly transmitted peer nonce, so a valid AUTH proves knowledge of the master key bound to the current session. The full exchange is shown in Figure 3.8: HELLO_T and AUTH_T travel on the GHz link, HELLO_R and AUTH_R on the MHz link. Both modules retransmit HELLO until they have seen the peer's HELLO, then send AUTH and wait for the peer's AUTH. Both handshake messages travel in plaintext, as the nonces are not secret and authenticity rests on the CMAC proof.

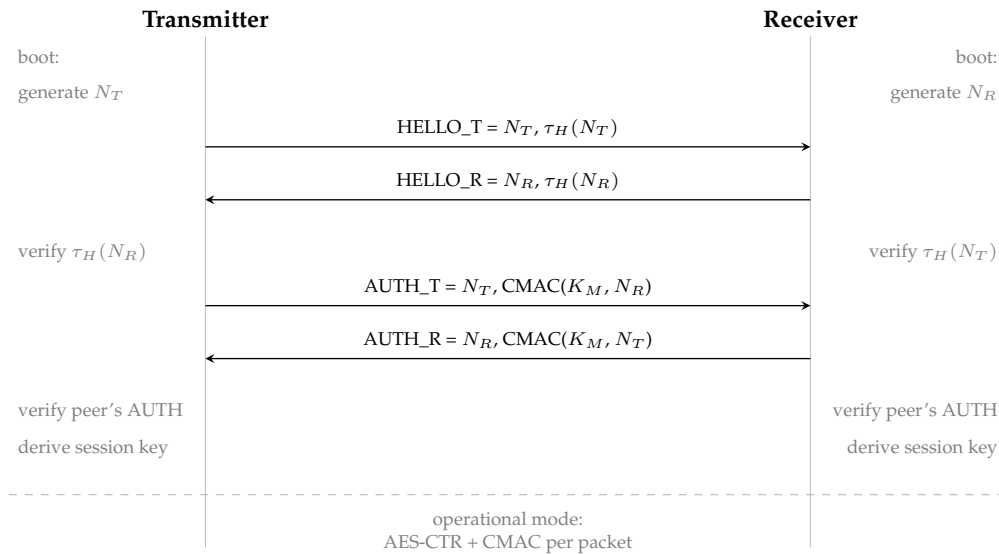


Figure 3.8: Handshake sequence between transmitter and receiver. HELLO carries a fresh nonce with a domain-separated MAC proof; AUTH carries a CMAC of the peer's nonce that proves knowledge of the master key. After both AUTH messages verify, both sides derive the same session key locally and switch to operational mode.

Session key derivation. Once both nonces are held and the peer's AUTH verifies, each module derives

$$K_S = \text{CMAC}(K_M, \min(N_T, N_R) \parallel \max(N_T, N_R)),$$

with byte-wise comparison. The canonical ordering matters because the two modules are otherwise symmetric and there is no explicit transmitter-or-receiver role in the key derivation. The construction is essentially a one-pass key derivation function with the master key as the input key, in the spirit of NIST

SP 800-108, and is secure assuming CMAC behaves as a pseudorandom function under AES. An attacker who exfiltrates K_M , can replay any past handshake and reconstruct every session key; full forward secrecy would require an ephemeral Diffie-Hellman exchange under the master key, which was deemed disproportionate for the threat model.

3.4.2 Per-Packet Encryption and Authentication

After the handshake, every transport-layer packet is processed through a four-step pipeline before transmission: (1) assemble the plaintext header (64-bit packet counter, 1-byte type, 7 reserved bytes, Section 3.3.2); (2) encrypt the payload with AES-128-CTR under K_S using the 128-bit IV equal to the packet counter in the low 8 bytes and zero in the high 8 bytes; (3) compute a 16-byte CMAC over the header concatenated with the ciphertext under K_S and truncate to 8 bytes; (4) concatenate header, ciphertext, and tag. This is the standard encrypt-then-MAC composition NIST recommends for combining a stream cipher with a separate MAC [6]. On the receiver the order is reversed: the tag is verified first, the packet is rejected on mismatch, and only then is the payload decrypted.

The 64-bit packet counter is monotonic within a session and rolls over only after 2^{64} packets, which at 50 packets per second (pps) corresponds to roughly 12 billion years; combined with the per-session key (Section 3.4.1), the CTR IV-uniqueness requirement is satisfied by construction. The same CMAC primitive authenticates each packet, the HELLO master-key proof, the AUTH tag, and the session-key derivation, so a single hardware accelerator covers all encryption and authentication paths. The 16-byte CMAC output is truncated to 8 bytes (NIST permits truncation down to 4 bytes for short-lived authenticators [7]), giving a forgery probability of 2^{-64} per attempt – at 50 pps a brute-force attacker would need $\sim 1.2 \times 10^{10}$ years of continuous transmission to expect one forgery. The CMAC is computed over the header and the ciphertext, so any tampering with counter, type, or reserved bytes is detected before decryption; this is also why a retransmitted packet (Section 3.3.2) needs its CMAC recomputed even though the ciphertext is unchanged.

3.4.3 Replay Protection

The 64-bit packet counter that serves as sequence number and AES-CTR IV also acts as the replay-protection token. The receiver remembers the highest counter accepted in the current session and rejects any packet whose counter is not strictly greater, with a small reordering window for retransmissions. A second, automatic defence comes from the session boundary: each new session derives a fresh K_S from new nonces, so packets recorded during one session decrypt to garbage under the next key and the CMAC check rejects them outright.

3.4.4 Peer Reboot Detection

When one module reboots while the other is still operational, the rebooted side starts again from counter zero with a fresh nonce, and the surviving side begins to reject every packet because the counters look like a replay. The session layer detects this with a single rule: if a HELLO arrives at a module that is already in operational mode, it compares the embedded nonce against the peer nonce of the current session; if they match, the HELLO is treated as a stray retransmission and ignored, otherwise the layer interprets this as evidence that the peer has rebooted, tears down the session, and returns both modules to the boot phase (Figure 3.7). The HELLO proof τ_H is verified before the comparison, so an attacker without K_M cannot drag the session into a reset loop with forged HELLOs.

3.4.5 Hardware AES Sharing under FreeRTOS

The single AES peripheral (Section 2.5.3) is shared between the GHz TX/RX tasks, the MHz TX/RX task, and the security module. Every access is wrapped in a single FreeRTOS mutex; the longest critical section is on the order of 10 μ s, three orders of magnitude below the on-air time of one FLRC packet, so the serialisation is invisible at the link level.

3.5 Application Layer

The application layer connects the CAN bus on the TX to a UART output on the RX and decides everything not strictly about getting bits across the air: which CAN frames to keep under load, how often to send a keepalive when the bus is idle, and how the two modules are supervised at runtime. It meets F-02 (CAN interface and processing) and shapes the latency budget of NF-06. The layer itself contains very little code: an ISR drops CAN frames into an in-RAM buffer, the GHz transmit task drains it when building a packet, and on the RX the same task that decrypts the packet (Section 3.4.2) prints the recovered CAN frames to a UART. Figure 3.9 shows the TX/RX components (top) and how the data block grows on the TX and shrinks back on the RX (bottom).

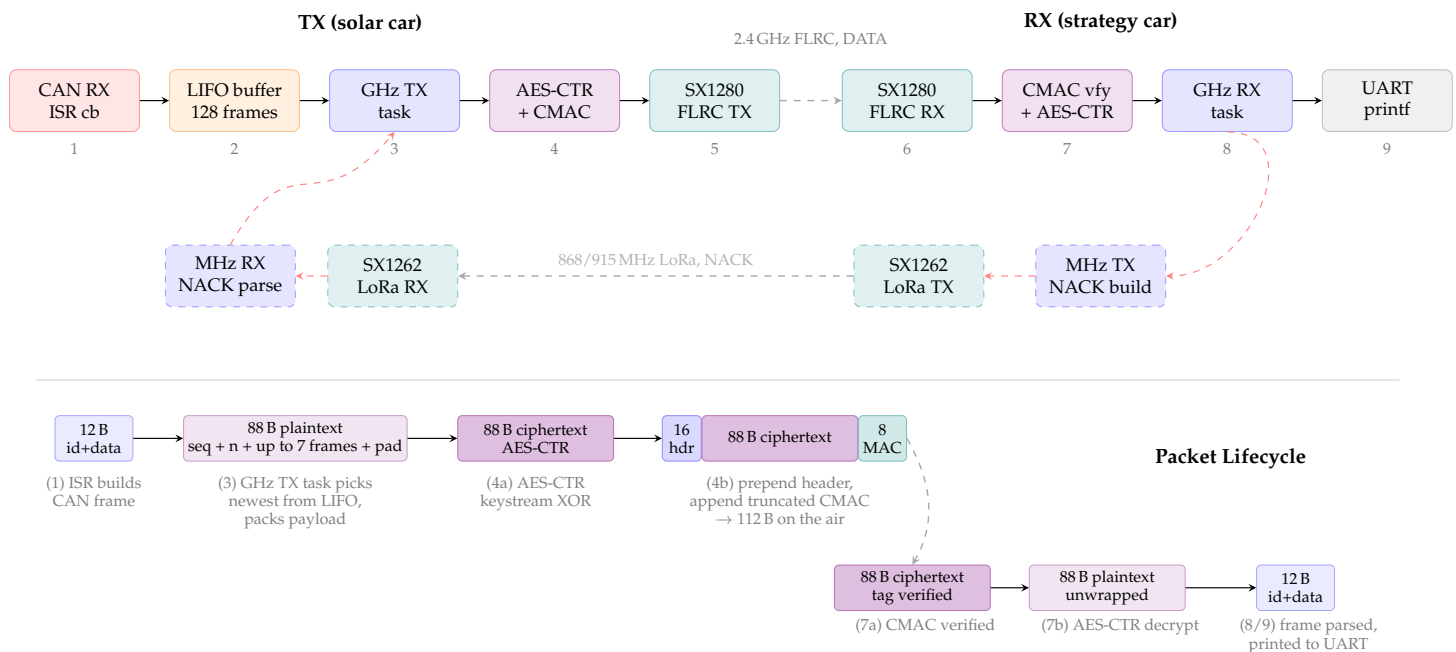


Figure 3.9: End-to-end path of one CAN frame through the system. Top: the chain of components on the TX (left) and RX (right), with the asymmetric MHz return channel for NACKs shown dashed. Bottom: how the same data block grows on the TX side (12 B raw frame, 88 B plaintext payload with sequence byte and frame count, 88 B AES-CTR ciphertext, 112 B fully framed packet with 16 B header and 8 B truncated CMAC) and shrinks back to a 12 B frame on the RX side.

3.5.1 Data Gathering

On the TX, every CAN frame that arrives on the on-board bus enters the system through a tiny Interrupt Service Routine (ISR) registered against the STM32L4A6 CAN peripheral. The ISR reformats the raw `(std_id, ext_id, ide, data[8])` tuple into a 12-byte `telem_can_frame_t` (Section 3.3.2) and pushes it into a single in-RAM buffer without touching FreeRTOS primitives or allocating memory, returning in well under a microsecond. The data path is therefore simply `ISR → buffer → transport layer`, with no intermediate worker task or queue. A small CAN ID filter sits in front of the buffer, accepting every frame by default but available to drop debug CAN IDs that would unnecessarily occupy radio bandwidth on race day; the filter is consulted from ISR context and is a constant-time table or range check.

3.5.2 Buffering

The buffer between the CAN bus and the radio absorbs the asymmetry between CAN traffic and FLRC throughput. Silvretta's 500 kbit/s CAN bus produces around 150 frames per second under typical race conditions, with bursts well above that during transients; the FLRC link carries up to 7 frames per packet at ~50 pps, so the system sustains roughly 350 fps over the radio. The buffer behaves as a stack rather than a queue: the GHz transmit task always takes the newest frames first, and any older frames that no longer fit are silently discarded. This is motivated by the nature of telemetry traffic, where most CAN messages are periodic state updates (battery SoC, motor rpm, etc.) and an older copy of the same signal carries no extra information; the freshest value always has the highest priority. The buffer is sized for 128 frames, which absorbs ~800 ms of typical traffic. Under sustained overload the system therefore degrades gracefully: even if the radio cannot keep up with a permanent CAN flood, the receiver still sees a continuously refreshed snapshot of the most recent state of the car at coarser granularity.

3.5.3 Keepalive and Run-State Supervision

When the CAN bus is silent (e.g. during a bench test), the transport layer would have no reason to transmit, but the receiver would then be unable to tell a healthy idle link from a dead one. The application layer therefore sends one `ALL_EMPTY` packet (Section 3.3.2) every second when no CAN data is queued; the receiver treats steady arrival as the link being up and a multi-second silence as a hint that something is wrong. On the RX side, the GHz receive task verifies the CMAC tag, decrypts the payload, and prints each recovered `(can_id, data[8])` tuple to the UART that connects to the Raspberry Pi and onward to the strategy network. The output is one human-readable line per frame, prefixed with the packet counter and the RSSI of the carrying packet; retransmissions are tagged as `RTX` versus the regular `R` so that the laptop-side parser can count true recoveries against fresh receptions. The same UART carries diagnostic output from the lower layers (handshake progress, NACK gaps, FLRC error counters).

Evaluation

This chapter presents the results of the three test categories introduced in Section 1.5: electrical verification, software testing, and field tests. Each section reports the measured values, compares them against the requirements from Table 1.1 and the theoretical predictions from Section 3.1.4, and notes any deviations.

4.1 Electrical Tests

Electrical testing begins with a continuity check on the bare PCB from JLCPCB before any soldering; with modern PCB manufacturers and CAD-aided design-rule checks, manufacturing shorts are rare. After hand soldering, a deliberate sequence of tests then finds shorts, bad solder joints, and component-level issues before any firmware is loaded, each step assuming the previous one has passed so that an issue is caught at the cheapest moment.

First power-up. The board is powered from a bench supply at ~ 5 V with a 100 mA current limit (preferred over USB-C because it caps current on a short). The 5 V rail feeds the DC/DC converter producing the 3.3 V digital rail. If the supply trips into current limiting, the joints are re-inspected under the microscope until the board draws a quiescent current of a few tens of milliamperes consistent with the Bill of Materials (BoM).

ST-Link connection. An ST-Link probe is connected to the SWD header. A successful connection simultaneously confirms a clean 3.3 V supply, correct reset and SWD routing, and an undamaged chip¹.

Peripheral bring-up. The firmware is flashed and dedicated routines exercise every peripheral in turn: the CAN chain (Peak CAN USB interface and matching scripts on both sides), the status LEDs blinked individually to verify polarity and soldering integrity, the two radio modules interrogated for their version register over SPI, the UART, USB-C voltage negotiation verified with an inline power tester, and the DIN switches toggled to confirm GPIO readback. Full integrated-system tests confirm that all subsystems interact correctly. With the radios active but the status LEDs and debug probe disconnected, the board draws around 15 mA from the 5 V rail (~ 75 mW), under the 0.1 W ceiling of NF-05; any significant deviation in current points to a partial short (high) or an unpowered component (low).

Thermal imaging. A thermal scan under typical load highlights any component running hotter than its neighbours, an early indicator of partial shorts or missing thermal pads. Figure 4.1 shows infrared and visible images of the assembled board; no component exceeded 30°C in 20°C ambient, with the MCU and the two RFICs as the warmest spots and no defects.

¹Wiring mistakes were caught here in both iterations: in V1 a boot pin was wired to the reset pin (fixed by bridging the RST button to ground via a 10 k Ω resistor), and in V2 the SWD UART3 RX pin was on an STM32 pin not capable of UART3 RX, harmless since the UART is simplex.

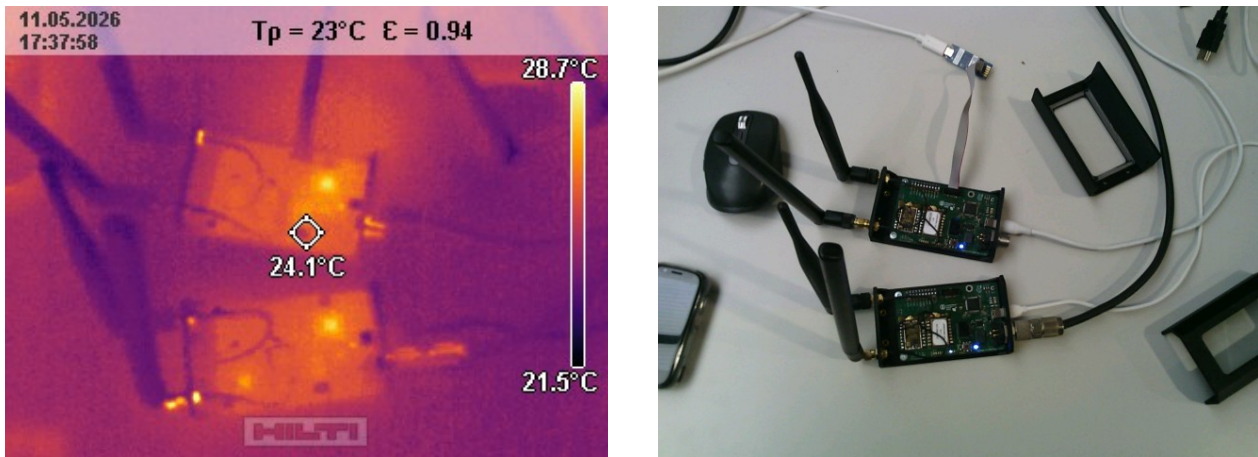


Figure 4.1: Thermal scan of the assembled board under typical load. Left: infrared image; right: visible-light image of the same scene. Entire board is relatively cool. Since one focus of the design is a low power consumption, components with high efficiency were chosen. Warmest components are the MCU and the Ghz RFIC TX side. The bright spots on the SMA connectors are reflections of body temperature in the glossy metal.

4.2 Software Tests

Software tests run on a fully assembled and electrically validated board (Section 4.1) and confirm that the firmware behaves as designed, that the RTOS leaves enough CPU and memory margin, and that the cryptographic code has no obvious defects.

Functional radio tests. The end-to-end radio path is tested in a controlled desktop setup with both boards on the same bench. A CAN injector publishes frames containing a monotonically increasing 64-bit counter; on the RX, a Python script parses each UART line into (radio counter, CAN counter, RSSI, retransmission flag) and verifies that every radio counter and every CAN message arrives exactly once. Any violation flags a defect in the retransmission logic, the LIFO buffer, or the session layer. The same harness is reused unchanged for the field tests in Section 4.3.

RTOS stack, task priorities, and memory. FreeRTOS is configured for static allocation only. Every task, queue, and timer uses `xTaskCreateStatic` and friends, and the C library's `malloc` is wired to fail. Stack health and per-task CPU usage are monitored with the XRTOS view in VS Code, which reads FreeRTOS kernel structures directly over the debug probe. Task priorities are set by observing the same view under load: the radio tasks run at the highest priority (a missed deadline causes packet-engine underrun), the CAN ISR runs even higher but outside the scheduler, and cryptographic and application tasks run lower. The final firmware shows no gaps in the radio packet counters and spends >65% of cycles in IDLE. The static RAM footprint is dominated by the 57 kB retransmission buffer and the task stacks, totalling ~130 kB of the 320 kB SRAM. Flash usage sits at ~180 kB of 1 MB, dominated by the vendor radio drivers and `printf`.

AI-assisted security review. The cryptographic core (Section 3.4) was reviewed with a large language model (Claude Opus) given the full source tree and the threat model of Section 2.5.1, asked to find any path by which a malicious radio packet could (a) cause memory corruption on the RX, or (b) translate into a CAN frame transmitted onto the on-board bus. The review walked every `memcpy` and variable-length loop on the radio ingest path and confirmed that all use compile-time-constant lengths guarded by explicit size checks, making classical stack/heap overflows unreachable from the radio. It also confirmed that the car-

side firmware contains no source-level path from radio bytes to a CAN transmit call, so the link cannot remotely actuate the car. The full review is reproduced in `documentation/security_analysis.md`.

4.3 Field Tests

Field tests close the loop between the theoretical link budget of Section 3.1.4 and the actual hardware in its intended environment, where range and data rate depend on the electrical quality of the board, the firmware, the antennas, the terrain, and the weather.

Method. Two laptops are connected, one to each telemetry board. The TX side pushes CAN frames containing a monotonically increasing 64-bit counter onto the on-board CAN bus; the RX side reads the serial output and logs every recovered frame with its radio counter, RSSI, and whether it arrived as a fresh reception or a retransmission. Two numbers are extracted offline: the raw radio packet loss (gaps in the radio counter), and the residual CAN frame loss after retransmission. The test is repeated at increasing distances in 5 m steps to characterise the link over its full operating range. The endpoints are static at each step; relative movement is simulated periodically by rapidly displacing and tilting the TX antenna.

Expected Results. The expected behaviour follows a clear progression: RSSI falls logarithmically with distance, which causes the radio to see more bit errors and therefore more raw packet loss. The retransmission mechanism absorbs this loss up to a point, CAN frame loss stays zero as long as the retransmission budget is not exhausted. Beyond a critical range, packet loss arrives faster than retransmissions can compensate, and CAN frames begin to drop at the application layer.

Site. The main field test took place on a former military airfield in Dübendorf, which offers a flat, open line of sight of up to 520 m and is close enough to the lab to allow several iterations per day. The RX antenna was mounted elevated on a movable tripod at 2.5 m above ground, while the TX antenna sat at vehicle height, matching the asymmetric height assumption used in the link budget of Section 3.1.4. Figure 4.2 shows the setup at the site. For most runs the TX module was placed under a spare solar-car canopy to test the TX strength through the actual epoxy the canopy is partly made of (DT121H from Delta-Preg, specially formulated for excellent RF transparency). The canopy had no measurable influence on the recorded RSSI.



Figure 4.2: Field test RX setup at the Dübendorf airfield. RX antenna is mounted 2.5 m from ground.

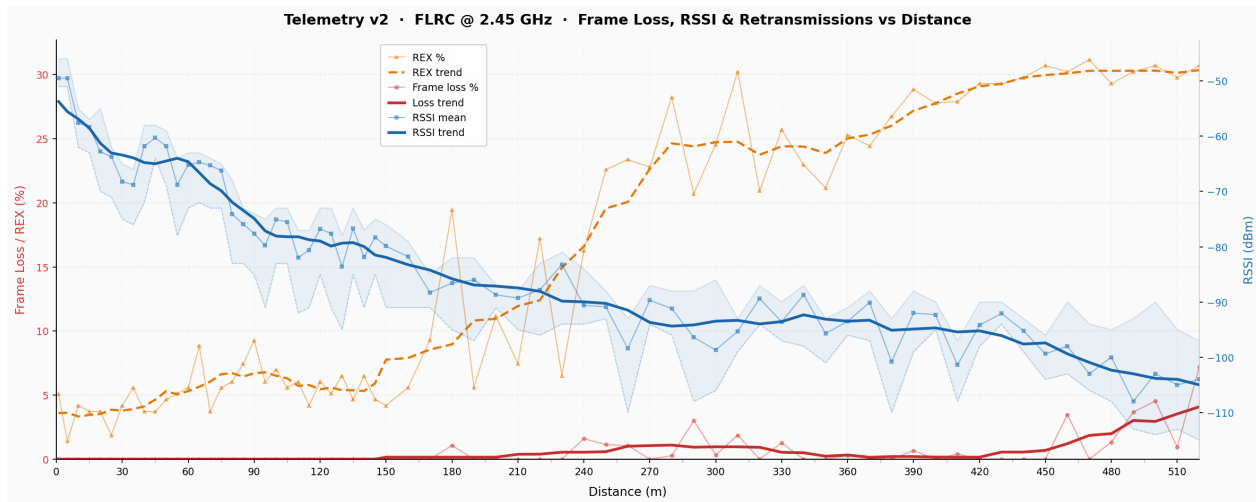


Figure 4.3: Visualisation of RX-side log from field test. 10 Seconds of 300 CAN fps was gathered, up to 150m, every 5m, after that, every 10m. The trace shows decreasing RSSI (blue), packet loss (orange) and resulting increasing frame loss (red). The retransmission mechanism is able to correct for up to 25 percent of packet loss, from where on actual frame loss happens. Low retransmission rates with still remaining non-correctable loss indicate fluctuations in link stability; the dips in min RSSI confirm that behaviour.

Results. The setup maintained a functional link up to approximately 520 m of line-of-sight separation. This measured edge sits within $\sim 6\%$ of the 552 m link-budget prediction of Section 3.1.4, validating the propagation model. At the test rate of 300 CAN frames per second, with seven frames packed into each radio packet, this corresponds to roughly 38 kbit/s of gross on-air throughput (approximately 28.8 kbit/s of CAN payload) on the 2.4GHz link. Up to 400 m, CAN frame loss was very rare and appeared random². This is well above the original requirements, especially considering that Dübendorf is a far-from-quiet RF environment and the link had to share the band with a considerable amount of nearby wireless infrastructure. At the NF-01 target distance of 500 m the raw packet loss was approximately 5 to 10 %, well below the 30 % requirement³. Figure 4.3 also shows the expected gradual logarithmic decrease of RSSI with distance; the resulting raw radio packet loss is absorbed by the retransmission mechanism up to roughly 25 % loss, beyond which non-correctable CAN frame loss begins to appear, consistent with the expected behaviour described above. At the far edge of the usable range, around 520 m, the residual CAN frame loss after retransmission rose to roughly 8 %, still well within the NF-01 budget and the figure reported in Table 5.1.

Three more subtle observations stand out. First, even at very short distances a few percent of raw packet loss remained, which the retransmission layer had to correct for, plausibly an artefact of the noisy spectrum at the airfield. Second, short intervals of degraded connection (at the 180 m mark, for example) proved most difficult to compensate for, as they deplete the retransmission budget within a single burst rather than distributing the loss over time. All figures above are reported as 10 s averages, since the underlying radio data is inherently noisy on shorter windows. Third, there appears to be localised interference around the 300 m mark, manifesting as several significant RSSI drops and a corresponding increase in packet and frame loss. Two plausible explanations exist. First, this may be a regime in which multipath components reflected off the concrete ground plane produce a significant destructive phase offset. Second, a test area for autonomous robots operates continuously in that area and appears to use some sort of Wi-Fi mesh networking, which

²During the race, the chase car carrying the RX unit will follow the solar car at a maximum separation of 50 meters.

³Remark on packet loss: as most CAN messages are published cyclically, packet loss has negligible effects on parameter modelling. The complete data set is additionally logged to an SD card on board the race car.

may contribute to the observed interference. The second explanation is more persuasive: the RSSI dips appear transient, as the maximum RSSI values within those windows do not exhibit a sustained decrease. The theory could be verified with a software-defined radio. Overall, the link met all requirements by a comfortable margin.

Conclusion

This thesis set out to design, implement, and characterise a CAN-to-RF telemetry unit that meets the constraints of solar racing: long range, low power, legal compliance under two regulatory regimes, and cryptographic protection of the on-board data. The resulting dual-link architecture, an STM32L4A6 driving an SX1262 LoRa downlink and an SX1280 FLRC uplink, satisfies all functional and non-functional requirements of Table 1.1.

Table 5.1: Functional and non-functional requirements for the CAN-to-RF telemetry unit and how they were met.

ID	Required	Achieved
F-01	Send CAN data wirelessly to the strategy car.	Dual-link RF relays CAN frames end-to-end.
F-02	Independent CAN interface and processing.	CAN ISR ingests and packetises all bus traffic.
F-03	Detect packet loss and retransmit.	Counter-gap detection with NACK retransmission.
F-04	Confidentiality, integrity, authenticity.	Per-packet AES-CTR + CMAC (encrypt-then-MAC).
NF-01	300 fps over a range of 500 m LoS with an effective frame loss of less than 30 %.	300 fps over a range of 520 m LoS with a max effective frame loss of 8 %
NF-02	The telemetry unit shall automatically re-establish connection when it was lost due to a reboot of unit.	Automatically re-establishing connection up to 500m after reboot of either device. Takes roughly one second (CAN data lost).
NF-03	The telemetry unit shall have a slim and lightweight form factor that fits within the aeroshell of the current solar car.	Final dimensions 106x68x28 mm with a total weight of 136 grams; mounts behind the RF-transparent fibreglass canopy window of the aeroshell (Figure 2).
NF-04	The system should use a hardware and software stack similar to other aCentauri modules to ensure compatibility.	Built on the shared STM32L4 MCU family, aCentauri PCB library, and FreeRTOS port; only the two radio RFICs and the DIN configuration switches differ from the aCentauri standard component library.
NF-05	The telemetry unit should consume less than 0.1 W in operating mode.	TX consumes roughly 0.06W, RX consumes roughly 0.08W
NF-06	The telemetry unit should introduce less than 1 second of latency.	Under 10ms of latency in normal conditions, LIFO buffer prevents worst possible latency.

Firmware and security. Layering the firmware along an adapted OSI model kept cryptographic, transport, and application concerns cleanly separated; the counter-gap detector, NACK polling, and circular retransmission buffer fit in well under a page of state machine and recover from realistic packet losses without involving the security layer. The encrypt-then-MAC construction over AES-CTR and AES-CMAC, anchored in a pre-shared master key and refreshed by a two-message HELLO/AUTH handshake on every boot, is simple to reason about and small to implement; replay protection and peer-reboot detection both fall out of the same monotonic counter and per-session key. The deliberately unusual LIFO application buffer is a good fit for cyclic CAN telemetry, where the freshest value is the one of interest. Fresh telemetry therefore reaches the receiver in well under 10 ms, satisfying NF-06; the only frames that can exceed one second are retransmitted packets.

Chip performance. The STM32L4A6 handles two radio links, an authenticated session, a 57 kB retransmission buffer, and the application logic without coming close to its limits. The hardware AES accelerator was the decisive feature: an 88-byte AES-CTR encryption plus CMAC tag completes in $\sim 30 \mu\text{s}$, three orders of magnitude below the FLRC on-air time. The implication for similar low-power IoT designs is that strong cryptography is essentially free when the chip exposes an AES peripheral that can be shared through a FreeRTOS mutex.

RF performance. All radio-side requirements were comfortably passed. The measured GHz range nevertheless falls short of the 2.3 km that Semtech quotes for the SX1280 at 260 kbit/s; that figure is reported without accompanying TX power, antenna gain, antenna height, or atmospheric condition, which makes it hard to reproduce. Consumer drone links such as DJI OcuSync push Mbit/s throughput to multi-kilometre range on 2.4 and 5.8 GHz, but rely on MIMO front-ends, beam-steering antennas, and a BoM one to two orders of magnitude above this board. Against the actual use case, a strategy car trailing the solar car at 50 meters, 38 kbit/s of on-air throughput out to 450 m on a 2.4 GHz omnidirectional link at the conducted TX power permitted in Europe and Australia is a good fit.

Comparison with a commercial unit. Table 5.2 sets the unit against the CANedge2 from CSS Electronics [5] (used by aCentauri in 2025). The comparison is not entirely like-for-like: the CANedge2 is a general-purpose two-channel logger that buffers to an SD card and offloads files over Wi-Fi to a local access point. Against the solar-racing use case the custom unit provides live transport over hundreds of meters at an order-of-magnitude lower power draw and price, at the cost of a single CAN channel and many additional features. The security models also differ: the CANedge2 relies on transport-layer protection (WPA2 on the Wi-Fi hop, TLS 1.2 to the server) plus optional AES-GCM log-file encryption at rest, whereas this work applies application-layer authenticated encryption to every frame over the air, with per-boot session keys and replay protection.

Table 5.2: Comparison of the custom telemetry unit with the CSS Electronics CANedge2 (manufacturer datasheet figures [5]). The CANedge2 price reflects list price; the custom unit’s price counts component cost at aCentaury’s conditions only and excludes labour.

	This work	CSS CANedge2
Wireless link	2.4 GHz FLRC + 868 MHz LoRa	2.4 GHz Wi-Fi (802.11 b/g/n) to local AP
Live link range	~520 m LoS	Wi-Fi range (tens of m), file offload only
Latency	<10 ms (1.2 s worst case)	Batched file upload (not real-time)
Power	~0.06–0.08 W	<1 W
CAN channels	1	2 (CAN FD)
On-board storage	via host	8–32 GB SD
Over-air security	Per-frame AES-CTR + AES-CMAC (encrypt-then-MAC), replay protection	WPA2 (Wi-Fi) + TLS 1.2 (transport)
Data-at-rest security	via host	Optional AES-GCM log files
Dimensions	106×68×28 mm	75×47×20 mm
Weight	136 g	100 g
Price	~55 CHF (sponsored parts)	~480 EUR

Limitations. The security protocol has not been peer-reviewed and should be treated as a student project rather than a hardened design. The field results rest on multiple days of static and pseudodynamic testing; more race representative data will become available once Silvretta, the current solar car, is driving. The exact race scenario is hard to reproduce on the ground.

Reflection. The work progressed in three hardware stages (Nucleo prototyping, first custom PCB, second iteration), with the firmware growing in step. The most demanding aspect was the inherent difficulty of debugging RF systems, where firmware, antennas, propagation environment, and regulatory configuration all collapse into a single observable: whether a packet arrives or not. The early investment in literature, link-budget, and regulatory analysis was a precondition for the timeline to hold once layout work had started. In hindsight, spanning RF design, custom hardware, firmware, and cryptography in a single project was arguably a touch ambitious for the scope of a Bachelor’s thesis.

Future Work

The telemetry unit covers the core CAN-to-RF link, but several extensions came up during the project that did not fit the timeline.

Tests on the solar car. All field tests were run with the unit on a movable tripod and bench, since Silvretta was not available during the testing window. Mounting the unit behind the aeroshell canopy and running it on the moving car would validate the link under real antenna placement, vehicle motion, and the high-voltage EMI environment.

Tests in a near-perfect environment. The Dübendorf airfield is electromagnetically noisy and, at under 600 m of usable line of sight, too short to exercise the LoRa downlink, whose theoretical range of 2 825 m (Section 3.1.4) was never approached. A quieter and more open site would let the GHz link be characterised without the ambient interference seen in Section 4.3 and the MHz downlink be measured against its much larger link budget.

FHSS reliability. The main open item on the radio side is the experimental FHSS mode of Section 3.1.3. The basic mechanism works, but keeping the TX and RX slot timers aligned under $\sim 5\%$ packet loss and drifting clocks is non-trivial. A dedicated low-power crystal or external time reference would relax this, and switching from FLRC to LoRa on the uplink would allow the SX1280's native FHSS support to be used directly. Together these would also unlock the full +20 dBm EIRP allowance and improve interference resilience.

GNSS and IMU on the TX. The RX currently relies on the host wall clock to stamp packets. A GNSS receiver on the TX would give every outgoing packet a precise on-board timestamp and absolute position; an IMU added alongside would enable on-board sensor fusion (as already used by the on-board logger) for a tighter, lower-latency state estimate.

Two-way CAN link. A bidirectional CAN-over-RF link would let strategy remotely adjust motor and inverter acceleration curves, push text messages to the cockpit display when the voice link fails, and wirelessly program the LED light show used at exhibitions. All three reuse the downlink that today carries only NACKs.

Second CAN interface and more MCU pins. Silvretta already has three CAN buses (a battery-internal BMU network, a solar-deck MPPT network, and the vehicle CAN); future cars may add a dedicated sensor CAN to separate sensor data from vehicle control commands, motivating a second CAN interface on the telemetry unit. The features above would not comfortably fit on the current STM32L4A6 (Section .3); an MCU variant with more pins would make extensions practical.

Attachments

.1 Software

[Download the software stack \(Google Drive\)](#)



Figure 1: QR code to a download link for the entire software stack running on the telemetry unit, including documentation.

.2 Low-Voltage CAN Modules of Silvretta

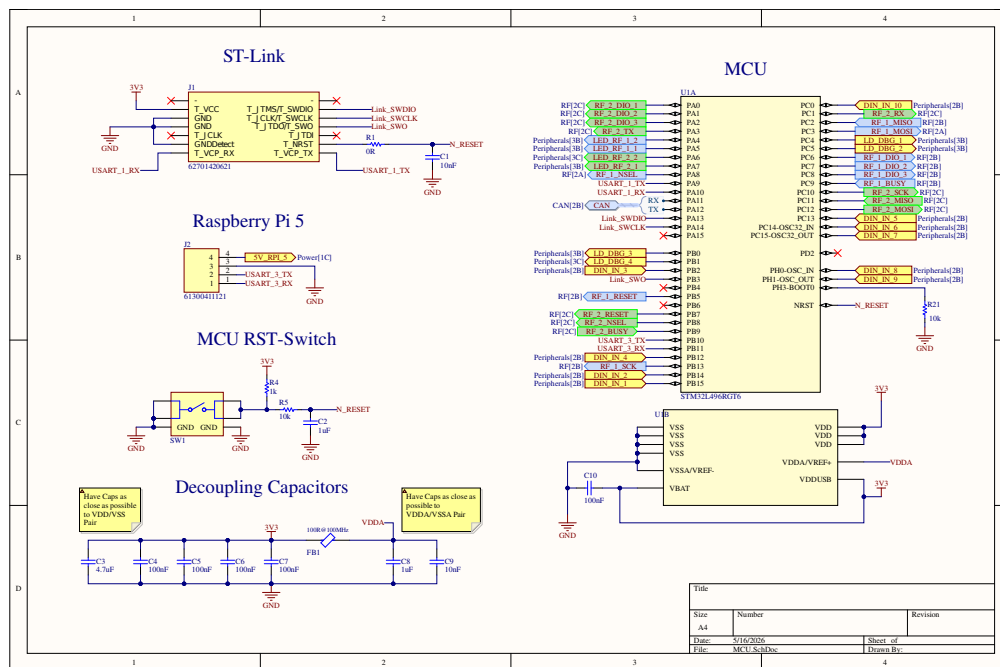
Table 1 lists the low-voltage modules connected to the on-board CAN bus of Silvretta. It complements Section 2.2 by giving the responsibilities of each node, so that the data sources the telemetry unit relays can be placed in context.

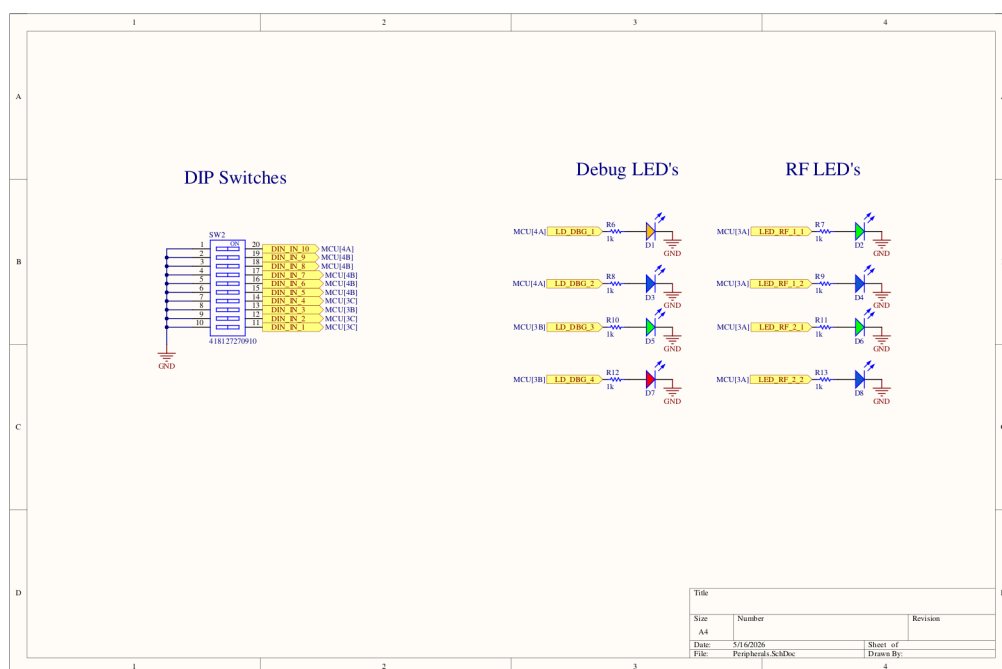
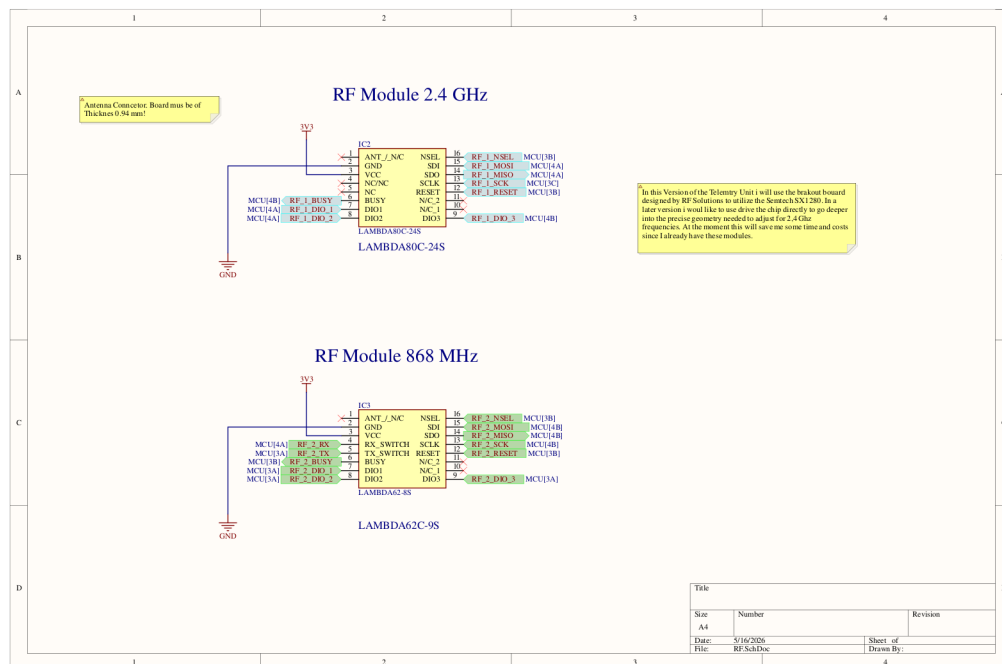
Table 1: Low-voltage modules on the on-board CAN bus of Silvretta and their main responsibilities.

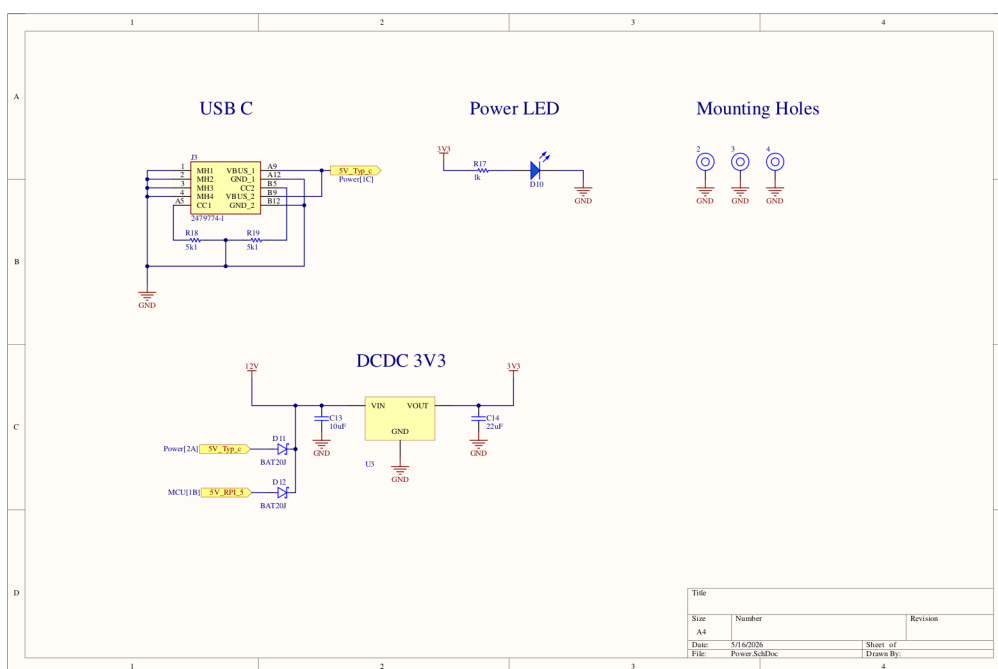
Module			Responsibilities
Vehicle (VCU)	Control	Unit	Manages 12 V peripherals (lighting, horn, side indicators, ventilation); reads the driver switchbox (reverse, high-voltage on, lights); feeds 12 V into the CAN bus.
Steering Wheel			Reads driver inputs (throttle, recuperation, indicators); computes motor commands for cruise control with a PID controller and sends them to the inverter; drives the E-Ink display (speed, cruise state, errors, power, photovoltaic input).
Logger			Logs all CAN traffic to SD card; reads a 9-axis IMU and an internal GNSS receiver; fuses both into an accurate position and tilt estimate of the car.
Telemetry (this work)			Gathers all incoming CAN messages, assembles them into encrypted packets, and transmits them to the strategy car.
Inverter (LV side)			Receives motor commands; publishes motor data such as RPM and power consumption.
Photovoltaic Disconnect			Can isolate the entire photovoltaic deck from the high-voltage system: supervises the Maximum Power Point Trackers (MPPTs).
MPPT (LV side)			Each MPPT drives a small group of solar cells to its optimal operating point to maximise harvested power.
Battery Management System (BMS)			Controls the Battery Management Units (BMUs), which in turn control the battery cells: publishes pack-wide data such as cell voltages and temperatures.

.3 Schematic of the Telemetry Board

The schematic of the second iteration of the telemetry board (Section 3.1.1) is reproduced on the following pages, exported directly from Altium Designer.







.4 Drawings

.4.1 Picture of Silvretta with Antenna Placement

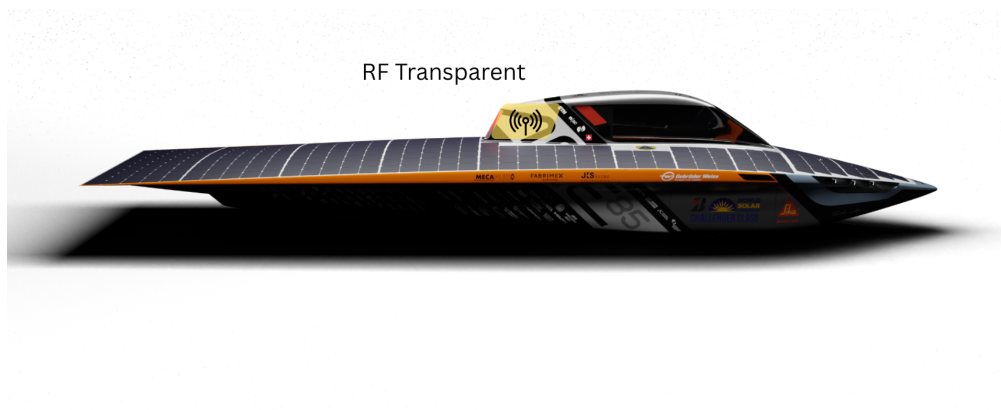


Figure 2: Rendered image of Silvretta with antenna placement and marking of the RF transparent fibreglass material.

.4.2 Link Budget Visualisation

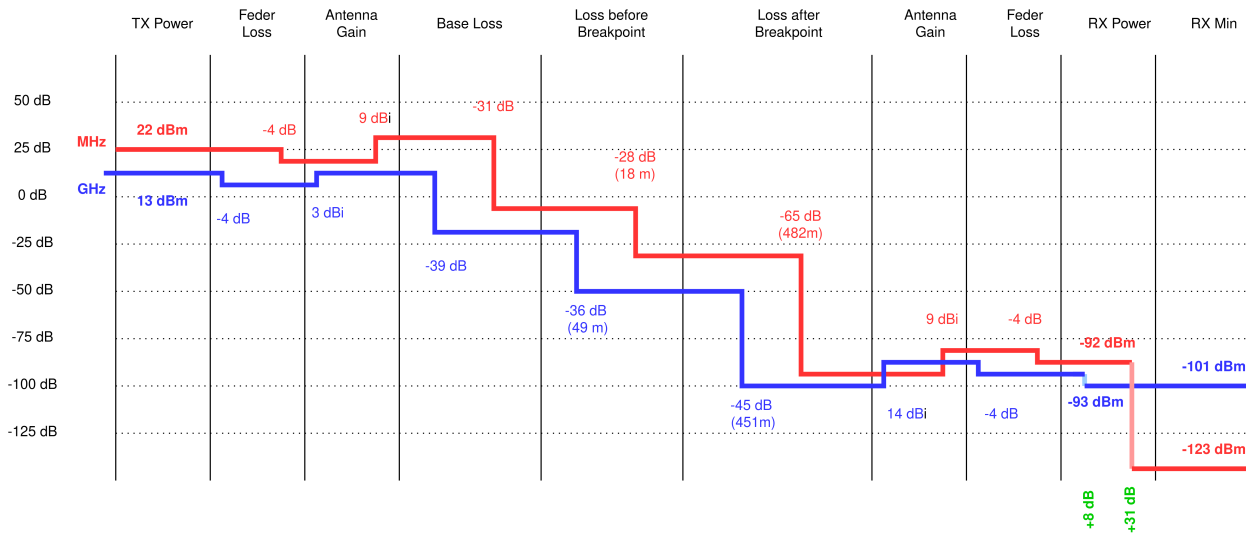


Figure 3: Link budget visualised over distance for both radios (Section 3.1.4). The difference between the received power and the receiver sensitivity shows the safety margin at 500 m range.

.4.3 Smith Chart

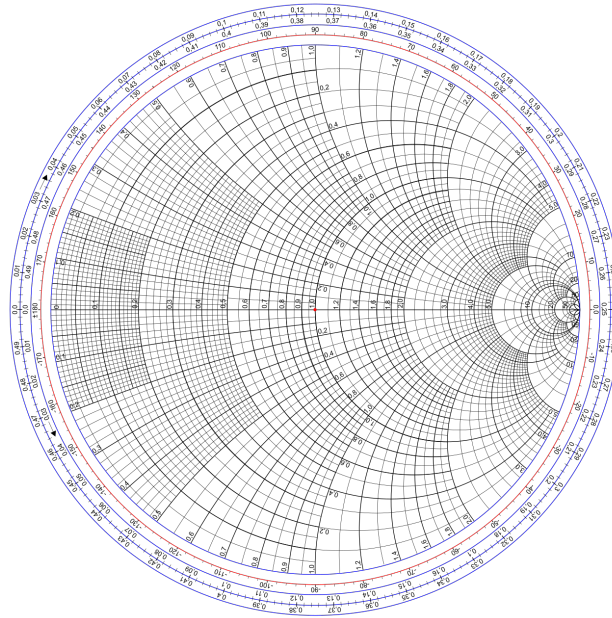


Figure 4: Smith chart, a polar plot of the reflection coefficient Γ overlaid with a coordinate system that lets the designer read complex impedance directly. Constant-resistance circles share the same real resistance, while constant-reactance arcs split into an inductive upper half ($+jX$) and a capacitive lower half ($-jX$). Used in this work together with a VNA to characterise and tune antennas and RF components (Section 2.4.2, Section 4.1).

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